MATHEMATICS GRADE 10: QUADRATIC EQUATIONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.3: Quadratic Expressions and Equations
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– Quadratic expressions: identifying, expanding, and simplifying expressions of the form $ax^2 + bx + c$ – Factorisation of quadratic expressions (including difference of two squares and perfect squares) – Forming quadratic equations from real-life contexts (e.g., area problems, Harambee fundraising targets) – Solving quadratic equations by factorisation – Solving quadratic equations by completing the square – Solving quadratic equations using the quadratic formula – Applications of quadratic equations in day-to-day situations
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify and write quadratic expressions of the form $ax^2 + bx + c$ in different situations, b) expand and simplify quadratic expressions using algebraic rules, c) factorise quadratic expressions including perfect squares and difference of two squares, d) form quadratic equations from real-life contexts such as area, fundraising, and motion, e) solve quadratic equations by factorisation, f) solve quadratic equations by completing the square, g) solve quadratic equations using the quadratic formula, h) apply quadratic equations to solve problems in day-to-day situations, i) appreciate the relevance of quadratic equations in modelling real-world phenomena.
Core Competencies	– Communication and Collaboration: The learner discusses with peers to expand, simplify, and factorise quadratic expressions, and shares solution strategies for equations derived from real-life Harambee fundraising and farming contexts. – Critical Thinking and Problem Solving: The learner analyses real-life situations (e.g., calculating the dimensions of a shamba plot in Kericho given its area) to form and solve quadratic equations, selecting the most efficient method among factorisation, completing the square, or the quadratic formula. – Digital Literacy: The learner uses calculators and digital devices to verify solutions to quadratic equations and explore graphical representations of quadratic functions. – Learning to Learn: The learner reflects on errors in algebraic manipulation and refines personal strategies for solving quadratic equations across multiple methods. – Self-Efficacy: The learner builds confidence by successfully modelling and solving progressively complex quadratic problems

	rooted in Kenyan everyday contexts.
Core Values	– Unity: The learner works collaboratively in groups to explore and compare different methods of solving quadratic equations, appreciating diverse approaches. – Responsibility: The learner takes responsibility for accuracy in algebraic working, understanding that errors in quadratic modelling (e.g., miscalculating a land area or fundraising shortfall) have real consequences. – Integrity: The learner honestly records and presents their working, showing all steps in solving quadratic equations rather than copying answers. – Social Justice: The learner recognises how mathematical literacy, including understanding quadratic relationships, empowers community members to make fair and informed decisions about resources and fundraising.
Science & Engineering Practices	– Asking Questions and Defining Problems: Students pose mathematical questions about what dimensions or quantities satisfy a given quadratic constraint (e.g., "What side length gives exactly 96 m² for the Harambee hall floor?"). – Developing and Using Models: Students build and revise algebraic models (quadratic equations) to represent real-life situations, updating their models as new constraints are introduced. – Using Mathematics and Computational Thinking: Students apply factorisation, completing the square, and the quadratic formula as computational tools to obtain exact solutions, and use calculators to verify numerical answers. – Constructing Explanations: Students explain why a quadratic equation may have two, one, or no real solutions, connecting algebraic reasoning to the physical sense of the context (e.g., rejecting negative lengths). – Engaging in Argument from Evidence: Students compare solution methods and justify which method is most efficient for a given quadratic equation, supporting their argument with worked examples.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Critical Thinking and Problem Solving): The learner applies quadratic equations to analyse and solve real-life problems such as determining optimal dimensions for a community hall or maximising a farming plot's area in the Rift Valley. – Financial Literacy: The learner uses quadratic equations to model fundraising scenarios, such as calculating contribution amounts or shortfalls in a Harambee event, developing awareness of financial planning. – Environmental Education: The learner connects quadratic area models to land use and conservation contexts, such as calculating the area of irrigation plots along Lake Victoria to avoid over-use of resources. – Citizenship: The learner appreciates how mathematics supports equitable community development decisions, such as fair allocation of land or Harambee funds.
Career Connections	– Architecture and Civil Engineering: Quadratic equations are used to calculate structural dimensions, load distributions, and areas of buildings — directly relevant to constructing community halls, schools, and housing in Nairobi and beyond. – Agriculture and Land Management: Farmers and land surveyors in the Rift Valley use quadratic area models to optimise field layouts, irrigation coverage, and crop yield planning. – Finance and Accounting: Financial analysts and accountants use quadratic and polynomial models to project revenue targets, model loan repayments, and analyse cost-profit relationships. – Physics and Engineering: Quadratic equations model projectile motion, enabling careers in mechanical engineering, aerospace, and sports science (e.g., analysing the trajectory of a javelin thrown by a Kenyan athlete). – Data Science and Information Technology: Programmers and data scientists use quadratic and higher-order functions in algorithm optimisation, machine learning cost functions, and computer graphics rendering.
Focus for Lessons	This unit develops learners' ability to work fluently with quadratic expressions and equations, moving from conceptual understanding of the algebraic structure $ax^2 + bx + c$ through to practical solution methods and real-world application. Using the anchoring context of a community Harambee fundraising challenge in a Kenyan village, learners discover that some real-life problems cannot be solved with linear equations alone, motivating the need for quadratic thinking. Across seven lessons, learners progress from recognising and forming quadratic expressions, through three distinct solution methods (factorisation, completing the

	square, and the quadratic formula), to critically applying their skills in authentic Kenyan contexts such as land area calculations, construction planning, and financial modelling.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems — like building the Harambee hall in Githurai? KICD KEY INQUIRY QUESTIONS: 1. How do quadratic expressions and equations differ from linear ones, and why are they necessary? 2. How can we use quadratic equations to model and solve problems in day-to-day activities?
Anchoring Phenomenon	A community in Githurai, Nairobi, is planning a Harambee to raise funds to build a new multipurpose hall. The organising committee knows the hall must have a rectangular floor with an area of exactly 120 m^2, and that the length must be 2 metres more than the width. When the committee tries to find the dimensions using simple arithmetic or a linear equation, they get stuck — no straightforward linear method gives them an answer. A local Form 4 student suggests the problem requires a quadratic equation. Students are shown the committee's hand-drawn sketch on a manila paper: a rectangle labelled "width = w" and "length = w + 2", with "Area = 120 m^2" written in the centre. The puzzle is immediately visible: how can you find a single unknown when it appears multiplied by itself? Students are asked to predict the width before any instruction is given, and most guesses are either wrong or arrived at by trial-and-error — revealing the limitation of their current tools and creating genuine need for the new mathematics of this unit.
Storyline Thread	Lesson 1 — PREDICT & ANCHOR PHENOMENON: Students examine the Githurai Harambee hall problem. They predict the hall dimensions using trial-and-error and discuss why linear equations fall short. The class co-constructs the equation $w(w + 2) = 120$, identifying it as a new type of equation. Students are introduced to the general form $ax^2 + bx + c = 0$ and begin recognising quadratic expressions in varied contexts. The Driving Question Board (DQB) is opened and initial "What we notice / What we wonder" cards are posted. Lesson 2 — EXPANDING & SIMPLIFYING QUADRATIC EXPRESSIONS: Students learn to expand products such as $(w + 2)(w + 3)$ and $(2x - 1)(3x + 4)$ using the distributive law and FOIL method. Supporting phenomena (the Kericho farm fencing problem) is introduced to show how real contexts generate quadratic expressions. Students simplify expressions and begin to see the structure of $ax^2 + bx + c$ emerging consistently. DQB is updated: "We now know how quadratic expressions are formed." Lesson 3 — FACTORISATION OF QUADRATIC EXPRESSIONS: Students reverse the expansion process to factorise quadratic expressions of the form $x^2 + bx + c$ and $ax^2 + bx + c$, including perfect square trinomials and difference of two squares. Using the Harambee hall equation ($w^2 + 2w - 120 = 0$), students practise factorising and identify factors that correspond to real dimensions. The Gikomba mandazi revenue problem is introduced as a factorisation challenge. DQB update: "Factorisation lets us 'undo' expansion to find hidden values." Lesson 4 — SOLVING QUADRATIC EQUATIONS BY FACTORISATION: Students formally solve quadratic equations by setting each factor equal to zero (Zero Product Property). They apply this to the Harambee hall problem, obtaining $w = 10$ and $w = -12$, then reason about which solution is physically valid. Multiple Kenyan-context equations are solved in pairs. Students revisit the DQB and refine their model: "A quadratic equation can have two solutions; context tells us which one to use." Lesson 5 — SOLVING BY COMPLETING THE SQUARE: Students are introduced to completing the square as a method for equations that do not factorise easily. Using the Nakuru school garden problem ($x^2 + ... = 48$), they practise transforming quadratic equations into the form $(x + p)^2 = q$ and solving. The geometric interpretation (literally completing a square diagram) is drawn to

	connect algebra to area. DQB update: "Completing the square always works, even when factorisation fails." Lesson 6 — THE QUADRATIC FORMULA & THE MOMBASA STONE PROBLEM: Students derive the quadratic formula by completing the square on the general form $ax^2 + bx + c = 0$. They apply it to the Mombasa projectile problem ($-5t^2 + 20t + 25 = 0$) and the mandazi pricing problem, comparing results to earlier methods. Students use calculators to verify solutions and discuss the discriminant ($b^2 - 4ac$) to determine the number of real solutions. DQB update: "The quadratic formula is a universal tool — it works for any quadratic equation." Lesson 7 — APPLICATION, MODEL BUILDING & CONSOLIDATION: Students return to the full Harambee hall scenario and present complete solutions using all three methods, justifying their preferred approach. New application problems (land area, fundraising revenue, construction) are solved in groups. Students construct a final cumulative model (poster or digital slide) mapping the journey from recognising quadratic expressions to solving equations in context. The DQB is formally closed — all initial "wonder" questions are answered. Students reflect on how quadratic thinking empowers community problem-solving.
Supporting Phenomena	– A maize farmer in Kericho wants to fence a rectangular plot along one side of a river (so only three sides need fencing). She has exactly 60 metres of wire and wants the enclosed area to be 400 m^2. Students discover that forming the area constraint produces a quadratic equation, and that it may have two possible solutions — only one of which is physically sensible. – A mandazi vendor at Gikomba Market in Nairobi prices her mandazi at Ksh x per piece. She notices that when she raises the price by Ksh 5, the number of mandazis she sells per day drops by 10, yet her total revenue stays at Ksh 2,000. Students form a quadratic equation to find the original price, encountering a real-life revenue model. – A stone is thrown vertically upward from the top of a building in Mombasa. Its height above the ground (in metres) after t seconds is given by $h = -5t^2 + 20t + 25$. Students use the quadratic formula to find when the stone hits the ground, connecting quadratic equations to motion in a physical context they can visualise. – A Nakuru school wants to create a rectangular garden inside a 10 m × 8 m courtyard, leaving a uniform path of width x metres around it. The garden area must be 48 m^2. Students model the situation as a quadratic equation in x, reinforcing how quadratic equations arise naturally from geometric constraints.

LESSON 1 (80 minutes): Anchoring the Phenomenon: The Githurai Harambee Hall Problem	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit phenomenon by anchoring student curiosity in a real Kenyan community planning problem, reveal the limitations of linear reasoning, and introduce the general form of a quadratic equation.
Knowledge	By the end of this lesson, the learner should be able to: (a) distinguish quadratic expressions from linear ones by identifying the squared term; (b) write the equation $w(w + 2) = 120$ in the general form $ax^2 + bx + c = 0$; (c) recognise quadratic expressions in varied real-life contexts.
Skills	By the end of this lesson, the learner should be able to: (a) use trial-and-error and systematic substitution to estimate solutions; (b) expand and rearrange a product of two linear expressions into standard quadratic form; (c) co-construct a class Driving Question Board (DQB) by articulating observations and questions about the phenomenon.
Attitudes	The learner appreciates that new mathematical tools are needed when familiar methods fall short, and values collaborative inquiry as a way to make sense of unfamiliar problems.
Key Inquiry Question	How do quadratic expressions and equations differ from linear ones, and why are they necessary?
Purpose in Storyline	This is the PREDICT & ANCHOR lesson. Students encounter the Githurai Harambee hall phenomenon for the first time, exhaust their current tools (arithmetic and linear equations), and feel the productive need for a new class of equations. The DQB is opened, capturing the unit's initial curiosities. Every subsequent lesson will return to this hall problem as the organising context.
Safety Notes	No physical hazards. Ensure psychological safety: explicitly tell students that wrong predictions are expected and valued — 'We are not here to be right yet; we are here to notice what our tools cannot do.'

B. LESSON OVERVIEW
Students are shown a hand-drawn manila-paper sketch of the Githurai community's proposed Harambee hall — a rectangle labelled w and (w + 2) with Area = 120 m² written in the centre. Working individually then in pairs, they attempt to find the width using arithmetic trial-and-error and then a linear equation, discovering that neither method works cleanly. The teacher guides the class to formalise their struggle into the equation $w(w + 2) = 120$, expands it to $w^2 + 2w - 120 = 0$, and introduces the general quadratic form $ax^2 + bx + c = 0$. Students identify a, b, and c, compare the equation to linear ones they know, and begin posting 'What we notice / What we wonder' cards on the class Driving Question Board. The lesson ends with students recording a refined prediction and the open question: "How do we actually solve this?"

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Video: Plinko Probability]	Each student receives a printed (or ARES-displayed) copy of the manila-paper sketch: a rectangle labelled 'width = w metres' and	1. DISPLAY the sketch on the board (printed manila sheet or ARES screen). Say verbatim: 'The Githurai community committee	PREDICT-THEN-COMPARE TABLE: Students generate a personal data table of (w, Area) pairs before any instruction. This	Observable indicator: Can the student compute $w \times (w + 2)$ correctly for at least 3 chosen values of w and record them

Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	'length = (w + 2) metres', with 'Area = 120 m²' written inside. Working silently and individually for 5 minutes, students attempt to find a value of w that satisfies the area condition using only arithmetic — no algebra yet. They record their guesses and the resulting area in a two-column table (Guess for w | Computed Area). After individual work, pairs share and compare tables for 3 minutes. The teacher then cold-calls 5 pairs to share one guess each, recording all attempts on the board in a whole-class trial-and-error table. Students observe that guesses converge on a value near 10 but that no one can justify why 10 works without checking, and that fractional or non-integer answers create confusion. The teacher then challenges students: 'Can you write ONE equation — using only the letter w — that describes this situation exactly?' Students attempt this in their exercise books for 4 minutes before sharing.	drew this sketch last week. They need YOUR help. Do NOT use any formula yet — just use your number sense. Try to find the width w.' 2. CIRCULATE silently for 5 minutes. Do NOT answer questions about method — redirect with: 'What does area of a rectangle mean to you?' 3. After 5 minutes, say: 'Stop. Compare your table with your partner. Who got closer to 120 m²? How do you know?' Allow 3 minutes of pair talk. 4. COLD-CALL exactly 5 pairs — alternate across the room (front-left, back-right, middle, front-right, back-left). Record every guess on the board without evaluation. 5. Ask the whole class: 'Looking at this table — what pattern do you notice as w increases?' WAIT 12 seconds. Cold-call 3 students. 6. Pose the written-equation challenge: 'Now write a single equation using w that captures this rectangle problem exactly.' WAIT 10 seconds before allowing any pair talk. 7. COLD-CALL 4 students to share their equations. Write all attempts on the board — including incorrect ones like '2w + 2 = 120' — without correcting yet.	strategy surfaces prior knowledge, exposes the trial-and-error limitation, and creates cognitive dissonance — the foundation for genuine need. The whole-class table aggregation makes the pattern visible to all learners simultaneously.	accurately? Listen for students who write a linear equation (e.g., 2w = 120 or w + 2 = 120) — these reveal a key misconception to address in Phase 3. Collect 3 exercise books at random to check trial-and-error tables before Phase 2 begins.
Observe Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	The teacher shows two contrasting equations on the board side by side: a familiar linear equation from a previous topic (e.g., 'A farmer in Kericho has a rectangular shamba. Length = w + 2. Perimeter = 28 m. Find w.') and the Githurai hall equation. Students work in groups of 4 to answer three structured questions on a task card: (Q1) Solve the	1. Write both problems on the board simultaneously. Say: 'I am giving you a familiar problem next to our hall problem. Work in your groups of four. You have 8 minutes. Answer all three questions on the task card.' 2. CIRCULATE and LISTEN for the key insight: 'The area problem has w times w — it becomes w-	CONTRASTING CASES: Placing a linear perimeter problem beside the quadratic area problem makes the structural difference visible without direct instruction. Students construct the contrast themselves, which deepens encoding. This strategy embeds NGSS Science & Engineering Practice 6 (Constructing Explanations) by asking students to articulate in	Observable indicator: Does the group's Q3 sentence contain the idea of 'multiplying w by itself' or 'w squared'? Groups whose Q3 sentences mention only 'two unknowns' or 'more variables' still hold a misconception — note these groups for targeted questioning in Phase 3. Check that all groups correctly solved the linear Kericho problem in Q1

📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Kericho perimeter problem — how many steps does it take? (Q2) Try to solve the Githurai area problem the same way — what goes wrong? (Q3) Write down in one sentence WHY the area problem is harder. Groups record answers on a half-sheet of paper. After 8 minutes of group work, two groups share their Q3 sentences with the class.	squared.' If a group reaches this, ask them: 'How would you say that in a mathematical sentence?' — do NOT confirm yet. 3. After 8 minutes, say: 'Pens down. Group 3 — read me your answer to Question 2.' WAIT 8 seconds. Then ask Group 7: 'Do you agree or disagree with Group 3? Why?' 4. Ask the whole class: 'What is the HIGHEST power of w in the Kericho problem? What is the HIGHEST power of w in the Githurai problem?' WAIT 12 seconds. Cold-call 2 students. Write their answers prominently: 'Linear: highest power = 1. Githurai hall: highest power = ?' — leave this as an open question for Phase 3.	writing WHY the methods differ.	(expected answer: w = 6 m).
Explain Phase 📹 **VIDEO:** [Video: Plinko Probability](#) Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	The teacher models the algebraic formalisation step by step, connecting each algebraic move back to the hall sketch. Students follow along in their exercise books. Step 1: Area = length × width → $120 = (w + 2) \times w$. Step 2: Expand → $120 = w^2 + 2w$. Step 3: Rearrange → $w^2 + 2w - 120 = 0$. Students then copy the general form $ax^2 + bx + c = 0$ and identify $a = 1$, $b = 2$, $c = -120$ for the hall equation. The teacher shows three further examples of quadratic expressions drawn from Kenyan contexts (a maize storage silo, a triangular plot in Kisumu, a matatu fuel-cost problem) and asks students to identify them as quadratic or not quadratic, justifying by citing the squared term. Students complete a 5-item individual 'Is it quadratic?' card and exchange with a partner for peer-marking using a displayed answer	1. Return to the hall sketch on the board. Say: 'Let us now write what the Githurai committee's problem is saying — in the language of algebra.' Point to the rectangle and write each step slowly. After writing $120 = (w + 2) \times w$, ask: 'Before I expand — what will happen when I multiply w by w?' WAIT 12 seconds. Cold-call 3 students. 2. Write $w \times w = w^2$ on the board and circle it in a different colour. Say: 'THIS is what makes this equation different from everything you have solved before. This single term — w-squared — is the reason we need new tools.' 3. Write the general form $ax^2 + bx + c = 0$. Say: 'This is called a quadratic equation in one variable. The word quadratic comes from the Latin word for square — because the variable is squared.' 4. Ask: 'For OUR hall equation, what are a, b, and c?' WAIT 10	ANCHOR-BACK FORMALISATION: Every algebraic step is narrated by pointing back to a physical part of the hall sketch (the width side, the length side, the interior area). This strategy, grounded in NGSS Practice 2 (Developing and Using Models), prevents algebra from feeling abstract by keeping it tethered to the physical context. The 'Is it quadratic?' card enacts NGSS Practice 8 (Obtaining, Evaluating, and Communicating Information) at a basic recognition level.	Observable indicator: On the 5-item 'Is it quadratic?' card, students should correctly identify at least 4 of 5 items. A score of 3 or below indicates the student cannot yet reliably identify the squared term as the defining feature. Collect all cards — use scores to form targeted groups for Lesson 2's factorisation activity. Also listen for students who say $a = 1$ for $w^2 + 2w - 120 = 0$ — this confirms they understand the implicit coefficient.

	key.	seconds. Cold-call one student for each value. 5. Display the three Kenyan context expressions one at a time. For each, ask: 'Quadratic or not quadratic? Raise your hand if you say quadratic.' Count hands visibly, then cold-call one student who disagrees with the majority to justify. 6. Distribute the 5-item 'Is it quadratic?' individual card. Allow 4 minutes. Then say: 'Exchange books with your partner. I will display the answers. Mark honestly — a wrong answer now is a gift.' Display answers and allow 1 minute peer-marking.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces the class Driving Question Board — a large sheet of manila paper or a section of the classroom wall divided into two columns: 'What we NOTICE about the hall problem' and 'What we WONDER (questions we still have)'. Each student receives two sticky notes (or small paper squares). On the first, they write one thing they noticed during today's lesson. On the second, they write one genuine question they still have. Students post their notes on the board, grouping similar ideas as they go. The class then does a 3-minute silent 'gallery walk' reading each other's contributions. The teacher reads aloud 4–5 wonder questions that will anchor the unit's upcoming lessons, circling them with a marker and saying: 'These are the questions that will drive our work for the next several lessons. We will come back to this board after every lesson and update it.'	1. Unveil the blank DQB. Say: 'This board belongs to you — it will track what this class figures out together over the coming lessons about quadratic equations and the Githurai hall. Every lesson, we will add to it.' 2. Distribute sticky notes. Say: 'Write NOTICE on one — something you are now sure about. Write WONDER on the other — a question you genuinely have. You have 3 minutes. Be honest.' WAIT and CIRCULATE — if a student is stuck, prompt: 'What confused you today?' or 'What surprised you?' 3. Ask students to come up row by row and post notes. Encourage them to read nearby notes and cluster similar ones. 4. Conduct a 3-minute silent gallery walk. Instruct: 'Read at least 5 notes that are not yours. Put a small tick on any wonder question you share.' 5. After the gallery walk, identify the 4–5 most-ticked wonder	DRIVING QUESTION BOARD (DQB) LAUNCH: The DQB is a metacognitive anchoring tool (rooted in Project-Based Learning and NGSS Practice 1 — Asking Questions). By making student curiosities public and persistent on the wall, it creates accountability to prior questions, validates diverse levels of understanding, and gives every lesson a clear motivational thread back to the community phenomenon.	Observable indicator: Does each student's NOTICE note contain a mathematically accurate observation (e.g., 'the equation has w^2', 'expanding gives three terms', 'a linear equation would not have w^2')? Does each WONDER note reflect genuine uncertainty about the mathematics or the phenomenon (e.g., 'How do we actually find the value of w?', 'Are there two answers?', 'What if the area was not 120?')? Vague or off-topic notes indicate a student who is not yet engaged with the phenomenon — follow up individually.

		questions. Read each aloud. Say: 'These are OUR driving questions. Watch how the board grows as we answer them — and as new ones appear.' 6. Photograph the board (or have the class prefect sketch it) for the ARES platform record.		
Model Building Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	As the opening model-building activity of the unit, students draw their own annotated sketch of the Githurai hall in their exercise books. The sketch must include: (1) the rectangle with labelled sides w and (w + 2); (2) the area equation w(w + 2) = 120 written inside; (3) the expanded standard form $w^2 + 2w - 120 = 0$ written below; (4) a label identifying a, b, and c; (5) a speech bubble from a stick-figure Form 4 student saying 'I know the equation — but I still cannot find w!' This model is explicitly described as 'Version 1 — incomplete' because students do not yet have a solution method. Students write at the bottom: 'What I still need to figure out: ___.' This unfinished model will be revised in every subsequent lesson as new solution methods are learned.	1. Say: 'For the last 10 minutes, you will build the first version of your personal model of this problem. It will be incomplete — that is perfectly correct for today. Open your exercise book to a clean page and title it: GITHURAI HALL MODEL — Version 1.' 2. Walk through the five required elements one at a time, giving students 1.5 minutes per element. Draw your own version on the board simultaneously. 3. After the speech bubble, ask: 'Why does our Form 4 student know the EQUATION but still cannot find w?' WAIT 12 seconds. Cold-call 2 students. Expected response: 'We do not yet know how to solve an equation with w-squared.' Affirm: 'Exactly. That is what this unit is for.' 4. Instruct students to write the 'What I still need to figure out' sentence. Say: 'This sentence is your personal learning goal for this unit. Keep it visible — we will answer it together.' 5. Close the lesson by pointing to the DQB and the model simultaneously: 'Today we opened the question. Every lesson from here is one more tool to answer it. The Githurai community is counting on us to figure this out.'	INCOMPLETE ANNOTATED MODEL: Beginning with a deliberately unfinished model (NGSS Practice 2 — Developing and Using Models) establishes that modelling is an iterative process, not a one-time product. The 'Version 1' label normalises incompleteness and frames all future lessons as model revisions rather than isolated topics, creating coherent unit-level storyline continuity.	Observable indicator: Does the student's sketch correctly show all five elements? Specifically, check that (3) the standard form is correctly written as $w^2 + 2w - 120 = 0$ (not $w^2 + 2w = 120$), and that (4) a = 1, b = 2, c = −120 are correctly labelled. A student who writes c = 120 (positive) has not yet understood the rearrangement step — flag for follow-up at the start of Lesson 2.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) PHENOMENON UPTAKE — Did students genuinely struggle with the trial-and-error phase, or did one or two students immediately produce the correct value of w = 10 by guessing? If the answer was reached too quickly, plan to introduce a more complex area value (e.g., 143 m^2) as an extension challenge in Lesson 2 to restore productive struggle. (2) LINEAR MISCONCEPTION — How many students wrote a linear equation (e.g., $2w + 2 = 120$ or $w + 2 = 60$) during Phase 1? This is the most common entry misconception. If more than one-third of the class held this error, open Lesson 2 with a 5-minute re-examination of the contrasting cases from Phase 2. (3) DQB QUALITY — Review the wonder questions posted. If most questions are procedural ('How do we solve it?') rather than conceptual ('Why does squaring happen?', 'Can there be two widths?'), prompt richer questioning at the start of Lesson 2 by asking: 'Could w have two different values? How would the community decide which one to use?' (4) MODEL COMPLETENESS — Check the Version 1 models in 5 exercise books before Lesson 2. Students who have not correctly rearranged to standard form need a brief individual correction at the start of the next lesson. (5) PACING NOTE — If Phase 4 (DQB) ran short due to time, students' NOTICE/WONDER notes can be completed as a homework task and posted at the start of Lesson 2 — do not skip the DQB, as it anchors the entire unit storyline.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed a hand-drawn sketch of the Githurai Harambee hall — a rectangle with width w, length (w + 2), and Area = 120 m^2 — and attempted to find w using trial-and-error arithmetic, generating a whole-class table of (w, Area) pairs.
What did I learn?	Students learned that multiplying two expressions each containing w produces a w^2 term, making the equation $w^2 + 2w - 120 = 0$ structurally different from linear equations; they identified this as a quadratic equation in the general form $ax^2 + bx + c = 0$ with $a = 1$, $b = 2$, $c = -120$.
How does this explain the phenomenon?	Students explained that linear methods fail for the hall problem because the area relationship forces w to be multiplied by itself, creating a squared term that cannot be isolated by simple arithmetic or one-step inverse operations — and that a new class of mathematical tools (quadratic equation methods) is needed to find the exact dimensions for the community's hall.

LESSON 2 (40 minutes): Expanding and Simplifying Quadratic Expressions	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will expand and simplify products of two linear expressions to show how quadratic expressions of the form $ax^2 + bx + c$ are formed, connecting this skill directly to the Githurai Harambee hall planning problem.
Knowledge	Students will know that multiplying two linear expressions using the distributive law or FOIL method always produces a quadratic expression of the form $ax^2 + bx + c$; they will identify the values of a, b, and c from the expanded form.
Skills	Students will be able to expand products such as $(w + 2)(w + 3)$ and $(2x - 1)(3x + 4)$ correctly, collect and simplify like terms, and represent real-world area problems as quadratic expressions.
Attitudes	Students will appreciate that algebraic expansion is a precise tool — not guesswork — that allows the community committee in Githurai to describe their hall problem mathematically and prepare to solve it in later lessons.
Key Inquiry Question	How do quadratic expressions and equations differ from linear ones, and why are they necessary?
Purpose in Storyline	Lesson 1 revealed the structure of the Harambee hall equation $w(w + 2) = 120$ but students could not yet work fluently with the left-hand side. This lesson equips them to expand and simplify any product of two linear expressions so they can rewrite the hall equation as $w^2 + 2w - 120 = 0$ with confidence — a prerequisite for factorisation in Lesson 3.
Safety Notes	No physical safety hazards. Ensure pair-work discussions are respectful; remind students that algebraic errors are a normal part of learning and should be shared openly so the class can learn from them together.

B. LESSON OVERVIEW
The lesson opens by returning to the Githurai Harambee hall sketch and asking students why the committee cannot simply expand $w(w + 2)$ by inspection without a systematic method. A supporting phenomenon — a farmer in Kericho fencing a rectangular tea-plot — is introduced to generate a second expansion problem in a fresh context. Through guided demonstration, pair practice, and a class gallery of expansions, students master the distributive law and FOIL method. By the end, every student has produced at least three correctly expanded and simplified quadratic expressions and can articulate how the $ax^2 + bx + c$ structure emerges. The DQB is updated to record the new evidence.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Area Model Multiplication Source: PhET Interactive	Students are shown the Githurai Harambee hall sketch from Lesson 1 (rectangle, width = w, length = w + 2, Area = 120 m²). The teacher writes $w(w + 2)$ on the board and asks: 'In your exercise book, write what you think $w(w + 2)$ equals	Display the hall sketch prominently. Say exactly: 'The committee in Githurai wrote w times the quantity w plus 2. Before I show you anything, I want your best mathematical guess — what does that product equal? Write it	Elicitation of prior knowledge through prediction; deliberate surfacing of misconceptions (especially conflating $w^2 + 2$ with $w^2 + 2w$) so they can be directly addressed during the Explain	Scan written predictions to identify the most common error. Note students who write $w^2 + 2w$ correctly — use them as peer explainers later. Use a show-of-hands poll: 'How many wrote an answer that includes w^2?' to gauge

Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	when you multiply it out — do not solve for w yet, just expand the brackets. Write your prediction in 60 seconds.' Students write individually, then share with a neighbour. Common predictions collected on the board include: $w^2 + 2$, $w^2 + 2w$, $2w^2$, and $w^2 + w^2$. The teacher does NOT evaluate yet — all predictions remain visible throughout the phase.	now.' Allow 60 seconds of silent individual writing. Then say: 'Share with the person next to you. You have 30 seconds.' Cold-call four students to share predictions and write each on the board labelled with the student name. Say: 'All of these are on the board. By the end of today, you will know which is correct and why.' --- WAIT TIME: After writing each prediction, pause 5 seconds before moving on to allow students to compare mentally.	phase.	baseline understanding.
Observe Phase 📹 **VIDEO:** Video: Area Model Multiplication Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces the Kericho tea-farm fencing problem on a hand-drawn manila card: 'Mwangi is fencing a rectangular tea-plot in Kericho. The length of the plot is $(x + 5)$ metres and the width is $(x + 3)$ metres. Write an expression for the area of the plot.' Students work in pairs for 3 minutes, attempting the multiplication using any method they know — some multiply term by term, others try repeated addition, others draw a box/grid. The teacher circulates and collects three distinct student approaches on the board (grid method, partial multiplying, FOIL attempt) without correction. Students observe: all approaches produce x^2, cross-terms, and a constant, but differ in organisation and completeness. The teacher highlights: 'Notice that whatever method you used, an x^2 term appeared. Why?'	Distribute or display the Kericho manila card. Say: 'Work with your partner. You have 3 minutes. Use any strategy you know to find an expression for the area.' Circulate actively — look for a student using a grid/box, a student multiplying each term one by one, and a student attempting FOIL. Say to each: 'May I put your work on the board? I want the class to see different approaches.' After 3 minutes, present the three approaches side by side. Say: 'Look at all three methods. What do they have in common? What is different?' --- WAIT TIME: After posing the comparison question, wait 10 seconds in silence before taking responses. If no response, say: 'Whisper your observation to your partner first.'	Multiple representation comparison: students see that different expansion methods all yield the same quadratic structure, building the concept that the $ax^2 + bx + c$ form is an inevitable outcome of multiplying two linear expressions, not a coincidence.	Circulate and note which pairs expand both brackets correctly versus which expand only the first term against the whole second bracket. Use this to plan the targeted example in the Explain phase.
Explain Phase 📹 **VIDEO:** Video: Area Model Multiplication Source: PhET Interactive Simulations	The teacher demonstrates the distributive law and FOIL method step by step using the hall equation first, then the Kericho problem, then two new examples. Step 1 — Revisit the hall: $w(w + 2) = w \times w + w \times 2 = w^2 + 2w$.	Use a two-colour chalk or marker system: one colour for the multiplication arrows, another for the simplified terms. Say: 'Watch the arrows — every term in the first bracket multiplies every term in the second bracket. No term is	Worked example with FOIL arrow labelling; peer-check of guided practice; explicit connection back to the phenomenon (the hall equation) to ensure expansion is not treated as an isolated skill; progressive complexity (monomial	Collect books of three target students (one high, one mid, one struggling) during the 90-second silent task to review working. Use exit-check question: 'What is the value of b in the expansion of $(2x - 1)(3x + 4)$?' Students write answer

(English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students check against their Predict-phase answer and see who was correct. Step 2 — FOIL for two binomials: teacher writes $(x + 5)(x + 3)$ and labels each arrow: First $(x \cdot x = x^2)$, Outer $(x \cdot 3 = 3x)$, Inner $(5 \cdot x = 5x)$, Last $(5 \cdot 3 = 15)$. Collect like terms: $x^2 + 8x + 15$. Step 3 — Students copy and complete two guided examples in their books: (a) $(w + 9)(w - 10)$ and (b) $(2x - 1)(3x + 4)$. Teacher circulates and gives corrective feedback in real time. Step 4 — Stress the structure: teacher draws a table mapping each expansion to a = ?, b = ?, c = ?: 'From $w^2 + 2w$ we now need to link this back to our hall. If Area = 120, we write $w^2 + 2w = 120$, or equivalently $w^2 + 2w - 120 = 0$. We now know what that left side means — it came from expanding.' Students annotate their books.	allowed to escape.' After Step 2, say: 'Try $(w + 9)(w - 10)$ on your own right now. I will give you 90 seconds of silence.' --- WAIT TIME: After issuing the 90-second task, stand still and do NOT prompt. After 90 seconds say: 'Swap books with your neighbour and check the first line of working.' For example (b), say: 'This one has coefficients in front of x. Does FOIL still work? Predict yes or no before you try.' Cold-call a student who predicted no, ask them to explain their reasoning, then work through the example to show FOIL still holds. End by saying: 'How does this connect back to the hall? Someone tell me in one sentence.' --- WAIT TIME: 8 seconds of silence before accepting a response.	× binomial to binomial × binomial to coefficients greater than 1).	on a slip of paper — correct answer is b = 5. Count slips to measure class understanding before moving to Model Building.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Area Model Multiplication Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students return to the DQB from Lesson 1. The teacher reads aloud the 'What we wonder' cards still unanswered. Students write a new card to add: 'We now know — multiplying two linear expressions always gives $ax^2 + bx + c$. We used this to rewrite the Githurai hall problem as $w^2 + 2w - 120 = 0$.' Students also write one new wonder card for what they still cannot do: several students generate cards such as 'How do we reverse this to find w?' and 'What if expansion does not give neat numbers?' These are pinned on the board for Lesson 3.	Point to the DQB and say: 'In Lesson 1 you asked how we find w. Today we learned something new. Write one card that starts with the words: We now know. Write one card that starts with: We still wonder.' Give 2 minutes for writing. Collect cards, read two aloud, and pin them. Say: 'The wonder cards about reversing expansion — that is exactly where we are going in Lesson 3.' --- WAIT TIME: After reading each card aloud, pause 5 seconds so students can connect it to their own thinking before moving on.	DQB as a living evidence tracker: students see their understanding progressing lesson by lesson and recognise that unanswered questions are legitimate mathematical goals, not failures.	Read the 'We still wonder' cards to diagnose what Lesson 3 must address. If no student raises the idea of reversing expansion (factorisation), prompt: 'We can build expressions — but can we ever undo that? Write a wonder card about that idea.'
Model	In groups of three, students solve	Assign groups deliberately (mix	Group modelling task in a new	Collect one group model per table

Building Phase 📹 **VIDEO:** [Video: Area Model Multiplication] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	a mini-modelling task: 'A farmer near Kisumu wants to fence two adjacent rectangular chicken coops side by side. Each coop has width x metres and length (x + 4) metres. Write and simplify an expression for the total area of both coops combined.' Students draw a sketch, write the expression for one coop, double it or add the two, expand, and simplify. They then complete the model table in their books: Expression before expansion | Expression after expansion | Values of a, b, c. Groups share answers and teacher facilitates a whole-class convergence. Final model written on the board: Total area = $2x^2 + 8x$. Students identify a = 2, b = 8, c = 0 and discuss why c = 0 (no constant area added). Teacher says: 'Today we built the tool. The Githurai committee now has the equation $w^2 + 2w - 120 = 0$ written in standard form. Next lesson we will reverse this process to find the hidden dimension.'	ability levels). Say: 'Your model must include a sketch, the unsimplified expression, and the simplified expression with a, b, c identified. You have 5 minutes.' Circulate and ask each group: 'Why did a squared term appear? Can you explain it in words without using algebra?' This probes conceptual understanding beyond procedural fluency. After groups share, draw the connection back to the hall: 'The hall equation $w^2 + 2w - 120 = 0$ is now fully formed using everything you practised today. What do you still need to do to help the Githurai committee?' --- WAIT TIME: After this final question, wait 10 seconds for student responses before summarising.	Kenya context (Kisumu poultry farming) that requires students to transfer the expansion skill; model table makes the $ax^2 + bx + c$ structure explicit and persistent; closing connection to the phenomenon maintains the narrative thread of the storyline.	at the end of the lesson. Check for: correct sketch, correct unsimplified expression, correct expansion, correct identification of a, b, c. Use these to inform the opening of Lesson 3 — groups with c errors will need targeted attention during factorisation.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did every student correctly identify at least one expanded quadratic expression with the right values of a, b, and c? (2) Were the connections back to the Githurai hall made explicitly enough that students could articulate why expansion matters for the community planning problem? (3) Which pairs or groups struggled with coefficients greater than 1 in example (b) — do they need a warm-up at the start of Lesson 3? (4) Did the DQB update feel owned by students, or did it feel teacher-directed? Adjust Lesson 3 opening accordingly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that multiplying (w)(w + 2) from the Githurai hall sketch and (x + 5)(x + 3) from the Kericho farm both produced expressions with a squared term, a middle term, and a constant — even when different methods were used.
What did I learn?	We learned that the distributive law and FOIL method are systematic ways to expand the product of any two linear expressions, always producing a quadratic expression of the form $ax^2 + bx + c$, and that the Harambee hall problem can now be written precisely as $w^2 + 2w - 120 = 0$.

How does this explain the phenomenon?	We can explain that the squared term in a quadratic expression arises because the variable appears in both brackets and multiplies itself during expansion; this is why the hall problem cannot be solved with a linear equation and why the committee in Githurai needs quadratic mathematics to find the dimensions of their hall.

LESSON 3 (80 minutes): Factorisation of Quadratic Expressions	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to factorise quadratic expressions of the form $x^2 + bx + c$ and $ax^2 + bx + c$, including perfect square trinomials and difference of two squares, and to apply factorisation to the Harambee hall problem and the Gikomba mandazi revenue problem.
Knowledge	Learners will know that: (a) factorisation is the reverse of expansion; (b) a quadratic expression $x^2 + bx + c$ can be factorised by finding two numbers that multiply to c and add to b; (c) expressions of the form $ax^2 + bx + c$ are factorised by splitting the middle term; (d) perfect square trinomials and difference of two squares follow special patterns; (e) the factors of the hall equation $w^2 + 2w - 120 = 0$ reveal the real dimensions of the Harambee hall.
Skills	Learners will be able to: (a) factorise quadratic expressions of the form $x^2 + bx + c$; (b) factorise quadratic expressions of the form $ax^2 + bx + c$ by splitting the middle term; (c) identify and apply perfect square trinomial and difference of two squares patterns; (d) connect factorised forms to real-world dimensions and quantities; (e) verify factorisation by expanding back.
Attitudes	Learners will: (a) appreciate that mathematical tools such as factorisation unlock solutions that simpler methods cannot reach; (b) show persistence when working through multi-step algebraic processes; (c) value collaboration by discussing and checking each other's factorisations.
Key Inquiry Question	How can we use quadratic equations to model and solve problems in day-to-day activities?
Purpose in Storyline	In Lesson 1, learners discovered the hall-planning problem cannot be solved linearly and expressed it as $w^2 + 2w - 120 = 0$. In Lesson 2, learners expanded brackets and understood how quadratic expressions are built. Now in Lesson 3, learners reverse that process — factorising — so they can 'undo' the multiplication and find the actual width of the Harambee hall. The Gikomba mandazi revenue context is introduced as a second factorisation challenge, broadening the community relevance of the skill.
Safety Notes	No physical hazards. Ensure pair/group discussions are conducted respectfully. Learners who struggled in Lesson 2 with expansion should be seated near stronger peers during group work. Remind learners that incorrect first attempts are part of the mathematical process — create a psychologically safe environment for error.

B. LESSON OVERVIEW

Learners build on their Lesson 2 expansion skills by reversing the process to factorise quadratic expressions. Starting from the Harambee hall equation $w^2 + 2w - 120 = 0$ established in Lesson 1, they discover that factorisation is the algebraic tool needed to find the hall's actual dimensions. The lesson moves from guided factorisation of simple trinomials ($x^2 + bx + c$) to the more demanding $ax^2 + bx + c$ form using the splitting-the-middle-term method, and includes special products (perfect square trinomials and difference of two squares). A second real-world context — a Gikomba market mandazi seller whose daily revenue is modelled by a quadratic — is introduced as a factorisation challenge. By lesson end, learners can factorise the hall equation and interpret both factors in context, updating the Driving Question Board with a new insight.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Polynomials intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the manila-paper sketch of the Harambee hall (reintroduced from Lesson 1): rectangle labelled 'width = w', 'length = w + 2', 'Area = 120 m²'. The equation $w^2 + 2w - 120 = 0$ is written on the board. Learners are asked to predict: 'Without solving, can you guess what two bracket expressions, when multiplied together, might equal $w^2 + 2w - 120$?' Each learner writes a prediction in their exercise book in the form (w + __)(w + __) = 0. Learners then share predictions with a shoulder partner.	Teacher reattaches the manila-paper hall sketch to the board and says: 'Last lesson we learned to expand brackets — to multiply out. Today we run the process backwards. Look at the hall equation we built together: $w^2 + 2w - 120 = 0$. I want you to predict — just predict — what two brackets multiplied together might give us this expression. Write your prediction now in the form (w + __)(w + __). You have 90 seconds.' [Allow 90 seconds of silent individual thinking — do NOT circulate yet.] After 90 seconds: 'Now whisper your prediction to your partner. Do you agree? Disagree? Why?' [Allow 60 seconds of pair talk.] Cold-call 3 pairs: 'Pair at the front — what did you predict and what was your reasoning?' Record all 3 predictions on the board without evaluating them. Say: 'We will return to these predictions at the end of Phase 3 to see who was closest — and more importantly, why.'	Activation of Prior Knowledge + Prediction Anchoring. By asking learners to predict the factored form before instruction, the teacher surfaces existing intuitions about the relationship between products and factors. Recording predictions publicly creates a reference point that generates cognitive tension — learners are motivated to resolve whether their guess was correct and why.	Observable indicator: Each learner has a written prediction in the form (w + __)(w + __) in their book. Teacher scans predictions during pair-share to identify: (a) learners who correctly guess $(w + 12)(w - 10)$ or close variants — these can be challenged further; (b) learners with no attempt or wildly inconsistent signs — these need closer monitoring during guided practice.
Observe Phase 📹 **VIDEO:** Polynomials intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners observe three worked examples on the board/chart paper, each connected to a Kenyan context: (1) Simple trinomial: A Kericho tea farmer's profit is modelled by $x^2 + 5x + 6$. Teacher demonstrates factorisation step-by-step, naming every micro-step. (2) Trinomial with negative constant: The hall equation $w^2 + 2w - 120$. Teacher demonstrates splitting the middle term using factor pairs of −120. (3) Difference of two squares: A Mombasa tiling contractor finds the area of a border as $x^2 - 49$. Teacher demonstrates the special pattern. After each example,	Teacher says: 'We are going to look at three examples. Your job is not just to copy — your job is to watch for the pattern. I will narrate every step out loud.' EXAMPLE 1 — $x^2 + 5x + 6$: Write on board. Say: 'I need two numbers that multiply to +6 AND add to +5. I will list factor pairs of 6: (1,6) — add to 7, no. (2,3) — add to 5, YES.' Write $(x + 2)(x + 3)$. Say: 'Watch me verify by expanding.' Expand back on the board. 'Does everyone see the expansion matches? Thumbs up if yes.' [Wait 5 seconds. Address any thumbs down immediately.] EXAMPLE 2 — $w^2 + 2w - 120$ (the hall	Worked Example Analysis with Narrated Thinking (Cognitive Apprenticeship). The teacher makes invisible algebraic reasoning visible by narrating every decision — particularly the factor-pair search. Connecting Example 2 directly to the hall equation and the learners' own Lesson 1 predictions creates meaning: this is not abstract procedure, it is the tool that solves the community's real problem. The 'one sentence' annotation after each example forces learners to encode the key condition (multiply to c, add to b) in their own words.	Observable indicator: During silent worksheet time, teacher circulates and checks 6–8 books. Look for: (a) correct identification of factor pairs for all $x^2 + bx + c$ items; (b) correct sign handling for negative constants; (c) correct application of difference-of-two-squares pattern. Learners who complete early and correctly are asked in a whisper: 'Can you write your own real-world context for one of these expressions?' Learners still on question 2 after 5 minutes receive a prompt card listing factor-pair steps.

	learners copy the worked example AND write one sentence: 'The factor pair I needed was ___ and ___ because they multiply to ___ and add to ___.' A short 6-question observation worksheet (3 $x^2 + bx + c$, 2 $ax^2 + bx + c$, 1 difference of two squares) is completed individually in silence.	equation): Say: 'This is our hall equation. Same process — two numbers that multiply to −120 and add to +2.' Pause and ask the class: 'The product is NEGATIVE 120. What does that tell us about the signs of our two numbers?' [WAIT 12 seconds.] Cold-call 2 learners. Narrate the factor-pair search: (−10, 12) → multiply to −120, add to +2. Write $(w + 12)(w - 10)$. Say: 'Now look at the predictions on the board. Which prediction was closest?' EXAMPLE 3 — $x^2 - 49$: Say: 'This one has no middle term. Notice $x^2 - 49 = x^2 - 7^2$. This is called the difference of two squares.' Write the pattern $a^2 - b^2 = (a+b)(a-b)$ on the board in a box. Verify by expanding. Distribute the 6-question observation worksheet. Say: 'Complete this individually and silently. You have 8 minutes.' [Circulate, do NOT give answers — place a small tick near correct working or a dot near lines that need review.]		
Explain Phase 📹 **VIDEO:** Polynomials intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of 4 to factorise three connected problems: (A) The Harambee hall equation: $w^2 + 2w - 120 = 0$. Learners factorise, then write: 'The possible widths from the factors are ___ m and ___ m. The physically meaningful width is ___ m because ___.' (B) The Gikomba mandazi challenge: A mandazi seller at Gikomba market, Mama Akinyi, finds her daily revenue R (in shillings) when she sells n trays is modelled by $R = 6n^2 + 11n - 10$. Learners factorise $6n^2 + 11n - 10$ using the splitting-the-middle-term method. (C) A perfect square trinomial: A square garden in Kisumu has area $x^2 + 14x + 49$ m².	Say: 'Now you will explain the mathematics — not just do it. Work in your groups of four. For each problem, you must: (1) show every step of the factorisation; (2) state in plain Swahili or English what the factors mean in context.' PROBLEM A: 'This is still our hall equation. Factorise $w^2 + 2w - 120 = 0$. Then answer: which value of w makes sense as the width of a real hall, and why does the other value not work?' [Allow 5 minutes. Circulate. After 3 minutes, if most groups are stuck on sign choice, ask the whole class without giving the answer: 'If w is a width in metres, can it be negative?' WAIT 10 seconds. Cold-call 1 learner.]	Contextualised Group Problem Solving with Forced Interpretation. Factorisation alone is a procedural skill; requiring learners to interpret the factors in context (rejecting the negative root for the hall width; identifying the side of the Kisumu garden) elevates the task to conceptual understanding. The Gikomba mandazi context grounds the $ax^2 + bx + c$ method in a recognisable Kenyan economic situation, making the splitting method feel purposeful rather than arbitrary. Returning to Phase 1 predictions closes the predict–explain loop, reinforcing metacognitive awareness.	Observable indicator: Each group's written work for Problem A must show $(w + 12)(w - 10) = 0$ with the conclusion 'w = 10 m (valid) or w = −12 m (rejected — negative length is impossible).' For Problem B, correct factorisation is $(2n + 5)(3n - 2)$. For Problem C, correct factorisation is $(x + 7)^2$. Any group that cannot articulate WHY negative roots are rejected is flagged for targeted questioning in Phase 4.

	Factorise to find the side length. Groups present findings to the class using a mini-whiteboard or exercise book held up.	PROBLEM B: 'Mama Akinyi at Gikomba needs to understand her revenue model. Factorise $6n^2 + 11n - 10$. Hint: multiply the coefficient of n^2 by the constant: $6 \times (-10) = -60$. Find two numbers that multiply to -60 and add to $+11$.' [Allow 6 minutes.] PROBLEM C: 'The Kisumu garden. Factorise $x^2 + 14x + 49$. What special pattern do you notice?' [Allow 3 minutes.] After group work: cold-call one group per problem to present. For each presentation ask the rest of the class: 'Do you agree? Does anyone have a different method that gives the same answer?' After all three presentations, return to the board predictions from Phase 1. Say: 'Look at our predictions. Now we know the correct factorisation is $(w + 12)(w - 10)$. Whose prediction was closest? What made the difference between a correct and incorrect prediction?' [WAIT 12 seconds. Cold-call 2 learners.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Polynomials intro] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Learners individually write one new sticky note (or a slip of paper) with: (a) one thing factorisation can now do that expansion could not do; (b) one new question that arose during the lesson. Notes are placed on the class Driving Question Board under two columns: 'What we can now do' and 'New questions we have.' Learners also update their personal learning journals with the DQB insight statement for Lesson 3: 'Factorisation lets us undo expansion to find hidden values.'	Say: 'Before we move to model-building, take 3 minutes to write your sticky note. Answer these two things: What can factorisation do that expansion could not? And what is one new question you now have about the hall problem or any other quadratic?' [Allow 3 minutes of silent individual writing.] Collect sticky notes and read 5–6 aloud to the class, placing them on the DQB. Narrate as you place them: 'This learner says factorisation helps us find the width. Excellent — that goes under What We Can Now Do.' For each 'new question' note, ask the class: 'Can anyone partially answer this, or shall we keep it as an open question for the next lesson?' Cold-call 2 learners	Driving Question Board as Living Knowledge Artefact. The DQB makes cumulative learning visible across the 7-lesson sequence. Physically placing new understanding next to prior questions shows learners that knowledge builds iteratively. The explicit connection to the hall committee's practical need ('they now have dimensions') affirms the real-world stakes of the mathematics and sustains motivation for the remaining lessons.	Observable indicator: Each learner has produced a sticky note with both a 'what I can now do' statement AND a new question. Quality indicator: the 'what I can now do' statement references factorisation (not just 'I learned something new'). Teacher notes any recurring misconceptions surfaced in the 'new questions' (e.g., 'Why did we reject −12?' appearing multiple times signals the class needs more work on contextual interpretation in Lesson 4).

		to respond to 1 question. Finally, point to the DQB and say: 'Our hall committee in Githurai is getting closer to their answer. We now know the width is 10 m and the length is 12 m. But a question remains on our board: How do we solve quadratics when factorisation is difficult or impossible? That is where Lesson 4 takes us.'		
Model Building Phase 📹 **VIDEO:** [Polynomials intro](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative hall model (begun in Lesson 1 and updated in Lesson 2). They add a new layer to their diagram: inside the rectangle, they now annotate the factorised equation $(w + 12)(w - 10) = 0$ alongside the expanded form $w^2 + 2w - 120 = 0$. They write the solved dimensions: width = 10 m, length = 12 m, and verify: $10 \times 12 = 120$ m² ✓. On the model, they draw a second smaller diagram alongside: Mama Akinyi's Gikomba mandazi context, labelling the factorised revenue expression $(2n + 5)(3n - 2)$. Learners write a two-sentence 'model caption' explaining what each factor represents in each context.	Say: 'Take out your hall model diagram from Lesson 1. We are going to update it with what we discovered today.' Write on the board the model update checklist: (1) Add the factorised form next to the expanded form. (2) Write the solved dimensions: w = 10 m, l = 12 m. (3) Verify the area. (4) Add the Gikomba mandazi diagram with its factorised expression. (5) Write a two-sentence caption.' [Allow 8 minutes. Circulate and check that the area verification $10 \times 12 = 120$ is present in every model — this is non-negotiable.] With 2 minutes remaining, ask learners to share their captions with a shoulder partner. Cold-call 2 learners to read their captions aloud. Affirm strong captions that connect the factors to real quantities. Close by saying: 'Our model is growing. We started with a sketch and an area. Today we added the algebraic tool that cracked it open — factorisation. In Lesson 4, we will find what to do when factorisation is not easy or not possible, and we will meet the quadratic formula. The hall committee is waiting.'	Cumulative Diagrammatic Modelling (Progressive Model Revision). Requiring learners to update the same diagram across multiple lessons creates a tangible record of conceptual growth. The side-by-side placement of expanded and factorised forms on the same diagram makes the inverse relationship between expansion and factorisation visually explicit. The area verification step ($10 \times 12 = 120$ ✓) embeds the habit of checking answers in context — a key mathematical practice.	Observable indicator: Every learner's model diagram contains: (a) the factorised form $(w + 12)(w - 10) = 0$; (b) the solved dimensions w = 10 m and l = 12 m; (c) the verification $10 \times 12 = 120$ m²; (d) the Gikomba mandazi factorised expression $(2n + 5)(3n - 2)$. Exit check: Teacher collects 5 randomly selected models as a quick exit-ticket scan. Any model missing the area verification is returned with a sticky-note prompt: 'Does your answer actually give 120 m²? Show the check.'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Were learners able to articulate WHY the negative root ($w = -12$) is rejected in the hall context — or did they mechanically eliminate it without understanding? If the majority could not explain this, plan a 5-minute re-engagement at the start of Lesson 4 using a simple context: 'Can a field in Rift Valley have a negative width?' (2) Did the splitting-the-middle-term method ($ax^2 + bx + c$) cause significant confusion for the Gikomba mandazi problem? If more than one-third of groups could not complete Problem B independently, consider introducing a 'factor grid' or 'AC method' scaffold card in Lesson 4 before moving to the quadratic formula. (3) Were the DQB sticky notes substantive (referencing factorisation specifically) or generic? Generic notes may indicate that learners are completing the task procedurally without genuine reflection — consider modelling a high-quality sticky note at the start of Lesson 4. (4) Did the Gikomba mandazi context feel authentic and engaging to learners, or did it feel forced? Learner energy and discussion quality during Problem B will give you a real-time signal. Consider asking 2–3 learners at the door as they leave: 'Which problem felt most real to you today — the hall or the mandazi?' Use their answers to calibrate contextual relevance for Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed three teacher-narrated worked examples of factorisation: (1) $x^2 + 5x + 6$ (Kericho tea farmer profit), (2) $w^2 + 2w - 120$ (the Harambee hall equation), and (3) $x^2 - 49$ (Mombasa tiling contractor — difference of two squares). They completed a 6-question individual worksheet applying these patterns.
What did I learn?	Learners learned that: (a) factorisation is the reverse of expansion; (b) to factorise $x^2 + bx + c$, find two numbers that multiply to c and add to b; (c) to factorise $ax^2 + bx + c$, use the splitting-the-middle-term method (multiply $a \times c$, find factor pair, split the middle term, factor by grouping); (d) perfect square trinomials and difference of two squares follow special patterns; (e) the Harambee hall equation $w^2 + 2w - 120 = 0$ factorises to $(w + 12)(w - 10) = 0$, giving real width $w = 10$ m and length 12 m; (f) negative roots are rejected when the variable represents a physical measurement.
How does this explain the phenomenon?	Learners explained their factorisation reasoning through three group problems: (A) the hall equation — factorised and interpreted, rejecting the negative root; (B) Mama Akinyi's Gikomba mandazi revenue model $6n^2 + 11n - 10$, factorised to $(2n + 5)(3n - 2)$; (C) the Kisumu garden perfect square $x^2 + 14x + 49 = (x + 7)^2$. They updated their cumulative hall model diagrams and posted DQB sticky notes anchored to the insight: 'Factorisation lets us undo expansion to find hidden values.'

LESSON 4 (40 minutes): Solving Quadratic Equations by Factorisation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students will solve quadratic equations by factorisation, applying the Zero Product Property, and use contextual reasoning to select the physically valid solution from two mathematical roots.
Knowledge	Students will know that: (a) if a product of two factors equals zero, then at least one factor must equal zero (Zero Product Property); (b) a quadratic equation solved by factorisation yields two solutions; (c) context determines which solution is valid when a variable represents a physical quantity.
Skills	Students will be able to: (a) rearrange a quadratic equation into the standard form ax squared plus bx plus c equals zero; (b) factorise the left-hand side into two linear factors; (c) set each factor equal to zero and solve for the variable; (d) evaluate both solutions against the real-world context and justify the accepted answer.
Attitudes	Students will appreciate that mathematics produces multiple answers and that critical thinking about context is as important as computational accuracy.
Key Inquiry Question	How can we use quadratic equations to model and solve problems in day-to-day activities?
Purpose in Storyline	This lesson is the pivot of the unit — students have spent three lessons building the tools (recognising, expanding, factorising) and now finally use those tools to fully solve the Harambee hall problem and answer the committee's original question. The two solutions w equals 10 and w equals negative 12 create a productive cognitive conflict that anchors the big idea: mathematics gives you answers, but community wisdom and physical reasoning tell you which answer to use.
Safety Notes	No physical safety concerns. Ensure students do not dismiss the negative root without reasoning — challenge any student who simply says it is wrong to explain why it cannot represent a width.

B. LESSON OVERVIEW

Learners begin by revisiting the factorised Harambee hall equation (w plus 12)(w minus 10) equals 0 from Lesson 3 and predicting what happens when a product equals zero. Through guided questioning, they discover the Zero Product Property. They then formally solve the hall equation, obtaining w equals negative 12 and w equals 10, and reason in pairs about which root the committee should use and why. Three additional Kenyan-context equations (a matatu fare problem, a Githurai plot fencing problem, and a mandazi revenue equation) are solved in pairs to consolidate the method. The lesson closes with a DQB update and a concise model statement.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Simple hypothesis testing]()	Students are shown the factorised equation from Lesson 3 written on the board: $(w + 12)(w - 10) = 0$. The teacher does NOT yet explain what to do. Students are asked	Write clearly on the board: $(w + 12)(w - 10) = 0$. Say: 'Before I teach you anything new today, I want you to stare at this equation for 60 seconds. Think about what	Anchoring prediction: students activate prior knowledge about multiplication and zero before formal instruction. Recording predictions publicly creates	Observe sticky notes and the poll tally. If fewer than 30 percent of students predict both values, note this as a gap to address explicitly in the Explain Phase. If most

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Product Estimation with Whole Numbers and Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	individually to write on a sticky note: What value or values of w could make this equation true? They have 2 minutes of silent thinking, then share with a shoulder partner. A class poll is taken: How many pairs predicted w = 10? How many predicted w = -12? How many predicted both? Predictions are recorded on the board without comment.	has to be true for two things multiplied together to give exactly zero. Write your prediction on your sticky note.' WAIT 90 SECONDS in silence — resist any urge to prompt. After partner sharing, poll the class with a show of hands and record tallies. Say: 'Interesting — we have different ideas. Let us find out who is right and why.' Do not confirm or deny any prediction yet.	accountability and a reference point for later confirmation.	students predict only w = 10 (the positive root), this reveals an intuitive bias toward positive numbers — a key misconception to surface.
Observe Phase 📹 **VIDEO:** Simple hypothesis testing Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Product Estimation with Whole Numbers and Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher poses a concrete numeric warm-up: 'If I tell you that A times B equals 0, and A equals 5, what must B be?' Students answer instantly: zero. Then: 'If A equals 0, what must B be?' Answer: any number works, but the product is zero. Then: 'If neither A nor B equals 0, can their product be zero?' Students discuss — no. This builds the Zero Product Property inductively. The teacher then shows two substitution tests on the board simultaneously: Test 1 — substitute w = 10 into (w + 12)(w - 10): (10 + 12)(10 - 10) = (22)(0) = 0. True. Test 2 — substitute w = -12 into (w + 12)(w - 10): (-12 + 12)(-12 - 10) = (0)(-22) = 0. True. Students observe both substitutions and record what they see in their notebooks. The teacher then invites students to articulate the pattern in their own words before it is formally named.	Work through the numeric warm-up with rapid-fire questions directed at individual students: 'Akinyi, if 5 times B equals zero, what is B?' After each answer, ask 'Why?' to press for reasoning, not just recall. When moving to the substitution tests, say: 'Watch what I do — do not copy yet. Just watch and think.' After both tests, say: 'Turn to your partner. In one sentence, explain what you just saw.' WAIT 60 SECONDS. Cold-call two pairs using different words. Then say: 'What you just described has a formal name. It is called the Zero Product Property. Write this in your notebooks: If P times Q equals zero, then P equals zero OR Q equals zero or both.' Write the property formally on the board and underline OR.	Inductive observation: students see the property work numerically before it is named, so the name becomes a label for something they already understand rather than a new rule to memorise.	Listen to partner explanations during the 60-second turn-and-talk. Target: students should say something equivalent to 'at least one of the factors must be zero.' If students say 'both must be zero,' this is a misconception — address it immediately with a counter-example: 'If (5)(0) = 0, did the 5 have to be zero?'
Explain Phase 📹 **VIDEO:** Simple hypothesis testing Source: Khan Academy (English - US curriculum)	With the Zero Product Property established, students now formally solve the Harambee hall equation step by step. The teacher writes the full working on one side of the board and students copy into notebooks. Step 1: (w + 12)(w -	For the hall equation, narrate each step aloud while writing: 'I am going to apply the Zero Product Property — each factor gets its own mini-equation.' After writing both mini-equations ask: 'How many solutions do we have?'	Worked example followed by graduated release: the hall problem is fully guided; Equation A is semi-guided (teacher circulates); Equations B and C are student-led with peer support. The Equation B twist (two valid roots) deliberately	Use a 3-2-1 quick check at the end of pair work: each student writes 3 steps of the method, 2 solutions they found for any equation, and 1 reason they accepted or rejected a root. Collect or scan before closing the phase. Target: all students can

🔍 Search ARES for similar videos 📖 **HTML:** Product Estimation with Whole Numbers and Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	10) = 0. Step 2: Set each factor to zero — w + 12 = 0 giving w = -12; w - 10 = 0 giving w = 10. Step 3: Two solutions — w = -12 or w = 10. Step 4: Context check — the width of a hall cannot be negative. Reject w = -12. Accept w = 10 m. Step 5: Length = w + 2 = 10 + 2 = 12 m. Verify: 10 times 12 = 120 square metres. The committee can build the hall. Students then solve three practice equations in pairs, each set in a Kenyan context. Equation A (Githurai plot): A rectangular plot has area 84 square metres and length 5 m more than width. Form and solve the equation. [w squared + 5w - 84 = 0, factors (w + 12)(w - 7) = 0, w = 7 m, length = 12 m.] Equation B (Matatu fare): A matatu operator finds that revenue R in hundreds of shillings satisfies R squared - 7R + 12 = 0. Find R. [(R - 3)(R - 4) = 0, R = 3 or R = 4; both are positive so both are mathematically valid — operator must use business context to choose.] Equation C (Mandazi): The number of mandazi n a mama mboga can sell satisfies 2n squared - 5n - 3 = 0. Find n. [Factorise: 2n squared - 6n + n - 3 = 0; 2n(n - 3) + 1(n - 3) = 0; (2n + 1)(n - 3) = 0; n = -0.5 or n = 3. Reject n = -0.5. Accept n = 3 dozen mandazi.]	Pause for choral response. Then say: 'Mathematics has given us two answers. Now I need you to think like the committee in Githurai. Can a hall have a negative width?' WAIT 20 SECONDS. Ask: 'Wanjiku, which solution does the committee use and why?' After confirmation, say: 'This is crucial — rejecting a solution is not guessing. It is reasoning from context. Write this in your notebooks as a rule.' For the pair practice, circulate actively. For Equation B (both roots positive), pause the class after 5 minutes and ask: 'This time both answers are positive. Does context still matter?' Prompt discussion about whether the operator wants to maximise revenue. For Equation C, ask a pair to present their splitting-the-middle step on the board to review Lesson 3 skills.	extends thinking beyond the simple reject-negative rule.	articulate the Zero Product Property step and at least one contextual rejection reason.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Simple hypothesis testing Source: Khan Academy (English	Students return to the class DQB. The teacher reads aloud the current DQB statement posted in Lesson 3: 'Factorisation lets us undo expansion to find hidden values.' Students are asked: Has todays lesson added something new to this understanding? Each student writes one new card. Suggested contributions include:	Stand at the DQB and say: 'Three lessons ago, the Githurai committee was stuck. They could not find the dimensions of the hall. What can you tell them today that you could not tell them on Day 1?' WAIT 30 SECONDS for quiet reflection. Then: 'Write your card — one sentence only. Make it something you would say to the	DQB as metacognitive anchor: connecting new learning explicitly to the phenomenon reinforces that the mathematics serves a real purpose. Voting on card quality builds evaluative thinking about mathematical communication.	Review the posted cards after class. Cards that only state the numerical answer (w = 10) without referencing the method or the reasoning process indicate surface-level understanding — these students need additional support in Lesson 5 when the context becomes more abstract.

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Product Estimation with Whole Numbers and Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	'A quadratic equation can have two solutions but context decides which one to use,' 'The Zero Product Property is the bridge between factorisation and solving,' and 'A negative answer is not always wrong — we must check the context.' Cards are posted under a new heading: What we now know — Lesson 4. The teacher reads three cards aloud and asks the class to vote (thumbs up or sideways) on whether each captures the big idea accurately.	committee if they called you right now.' After posting, read three cards and model how to assess them: 'This card says the hall is 10 metres wide and 12 metres long — that is a fact, but does it explain the method? This card says we used the Zero Product Property — that explains how we got there. Which is more useful for the DQB?' Guide students toward process-level rather than answer-level statements.		
Model Building Phase 📹 **VIDEO:** Simple hypothesis testing Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Product Estimation with Whole Numbers and Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students individually draw a solution flowchart in their notebooks titled: How to solve a quadratic equation by factorisation. The flowchart must include five boxes: Box 1 — Write the equation in standard form ax squared + bx + c = 0. Box 2 — Factorise the left-hand side into two brackets. Box 3 — Apply the Zero Product Property: set each bracket equal to zero. Box 4 — Solve each mini-equation to get two solutions. Box 5 — Check context: reject any solution that is physically impossible. Below the flowchart, students write one sentence connecting this method back to the Harambee hall: 'Using this method, the committee in Githurai now knows the hall should be 10 m wide and 12 m long, with a verified area of 120 square metres.' Students compare flowcharts with a partner and add any missing step in a different colour. The teacher selects two contrasting flowcharts (one complete, one missing the context-check step) and displays them under a visualiser or on the board for a brief whole-class discussion about what is missing and why it matters.	Say: 'Before you close your notebooks, I want you to build a tool you can use in Lessons 5, 6, and 7. Draw the flowchart — it should take you 5 minutes.' Circulate and look specifically for the context-check box — this is the most commonly omitted step. When displaying the two contrasting flowcharts, ask: 'What does this flowchart do well? What is missing? Why does the missing step matter?' Do NOT identify the owner of either flowchart — anonymity keeps the focus on the mathematics, not the student. Close by saying: 'In our next lesson, we will meet equations that refuse to factorise. Our flowchart will need a new branch. Keep it — you will add to it.'	Model building as consolidation: the flowchart externalises the students procedural understanding and reveals gaps (missing context-check) that can be addressed. Comparing flowcharts models peer review as a mathematical practice.	Collect three to five flowcharts at random as exit tickets. Assess against four criteria: standard form step present, factorisation step present, Zero Product Property step named, context-check step present. Results inform the opening of Lesson 5 — if more than 40 percent of students omit the context-check, begin Lesson 5 with a quick re-engagement scenario before introducing completing the square.

D. TEACHER REFLECTION
After the lesson, ask yourself: (1) Did the Equation B twist (two positive roots) generate genuine discussion about context, or did students ignore it? If ignored, plan a brief re-engagement in Lesson 5. (2) Were students able to articulate the Zero Product Property in their own words, or were they only reciting the formula? (3) Did the DQB cards show process-level thinking (method, reasoning) or only answer-level thinking ($w = 10$)? (4) Do the flowcharts include the context-check step? (5) Are there students who are still relying on trial-and-error rather than the formal method? Identify them for targeted support during Lesson 5 pair work.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that setting each factor of $(w + 12)(w - 10) = 0$ equal to zero gives $w = -12$ and $w = 10$. When we substituted both values back, both made the equation true. We also observed that in the Githurai hall problem, only $w = 10$ makes sense as a physical width.
What did I learn?	We learned the Zero Product Property: if two factors multiply to give zero, at least one factor must equal zero. We learned that a quadratic equation solved by factorisation always produces two solutions, and that context — not just algebra — determines which solution to accept.
How does this explain the phenomenon?	We explained that the Harambee hall in Githurai must be 10 m wide and 12 m long because the Zero Product Property gives us two roots ($w = 10$ and $w = -12$) but a negative width is physically impossible. We also explained that sometimes both roots are positive and we must use deeper contextual reasoning to choose between them.

LESSON 5 (80 minutes): Solving Quadratic Equations by Completing the Square

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To equip learners with the method of completing the square as a universal technique for solving quadratic equations — including those that cannot be factorised — so they can find unknown dimensions and quantities needed for real community planning problems like the Githurai Harambee hall.
Knowledge	Learners will know that: (a) completing the square is a method of rewriting a quadratic equation in the form $(x + p)^2 = q$; (b) this method works for all quadratic equations, including those that do not factorise over integers; (c) the geometric interpretation of completing the square connects algebraic manipulation to area models; (d) the method produces exact solutions and is the algebraic foundation for the quadratic formula introduced in Lesson 6.
Skills	By the end of the lesson, the learner should be able to: (a) rewrite a quadratic expression $x^2 + bx + c$ in completed-square form $(x + p)^2 = q$ by halving the coefficient of x and squaring; (b) solve quadratic equations using the completed-square form; (c) draw and interpret the geometric square-completion diagram to justify each algebraic step; (d) apply completing the square to real-world dimension problems set in Kenyan contexts.
Attitudes	Learners will: (a) appreciate that mathematics offers systematic methods that go beyond trial-and-error; (b) show persistence when algebraic manipulation requires multiple steps; (c) value the geometric reasoning that underlies abstract algebra; (d) collaborate respectfully when verifying each other's completed-square workings.
Key Inquiry Question	How can we use quadratic equations to model and solve problems in day-to-day activities?
Purpose in Storyline	In Lessons 1–4, learners built the phenomenon, expanded and factorised quadratic expressions, and solved equations by factorisation. However, the Githurai hall equation $w^2 + 2w - 120 = 0$ factorises neatly — leaving learners wondering: what happens when an equation refuses to factorise? In Lesson 5, a new Kenyan context (a school garden in Nakuru) presents exactly such an equation. Learners discover that completing the square is a reliable, always-applicable method — and they update the Driving Question Board with this powerful insight, moving the class one step closer to solving any quadratic the Harambee committee might encounter.
Safety Notes	No hazardous materials are used. If square tiles or cut-out paper squares are employed in the geometric activity, ensure scissors are handled responsibly and that cut pieces are not left on the floor as a slip hazard.

B. LESSON OVERVIEW

Learners begin by encountering a quadratic equation from a Nakuru school garden context ($x^2 + 6x = 48$) that does not factorise easily over integers, exposing the limitation of Lesson 4's factorisation method. Through a guided geometric activity — physically arranging and completing a square diagram on grid paper — learners discover the completing-the-square algorithm. They practise transforming equations into the form $(x + p)^2 = q$, solve for x, and connect every step back to the Githurai hall phenomenon. The lesson closes with a DQB update and a cumulative model revision showing that completing the square generalises what factorisation could only do in special cases.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Comparing with multiplication: magic Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Square Roots and Irrational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a new scenario card (printed or displayed on the ARES platform): 'A school garden in Nakuru is rectangular. Its area is 48 m². The length is 6 m more than the width. Find the width.' Learners write the equation in their exercise books, attempt to factorise it, and record their prediction of the width. They also answer: 'Does this equation factorise as neatly as the Githurai hall equation?' Most learners will struggle to find integer factor pairs of −48 that sum to 6, surfacing the limitation. Learners share predictions with a partner using Think-Pair-Share.	Display the Nakuru garden scenario card and read it aloud together with learners. Say: 'Before I teach you anything new, I want you to write the equation this problem gives us — just like we did for the Githurai hall — and try to factorise it. You have 4 minutes. Go.' Circulate silently for 4 minutes, noting on a clipboard which learners write $x^2 + 6x - 48 = 0$ correctly and which attempt incorrect factor pairs. After 4 minutes, cold-call 3 learners to share their factor-pair attempts: 'Aisha, what factor pair did you try?' — WAIT 10 seconds. 'Brian, did you get the same pair?' — WAIT 10 seconds. 'Wanjiku, could you find integers that work?' — WAIT 12 seconds. Affirm the struggle without giving the answer: 'Notice that none of us can find a clean integer pair. This is NOT a mistake — this equation genuinely does not factorise over integers. That is exactly why we need today's lesson.' Point to the DQB and say: 'Our committee in Githurai will meet equations like this too. Let us find a method that never fails.'	Activation of Prior Knowledge + Productive Struggle: Learners deliberately apply the factorisation method from Lesson 4 to an equation it cannot solve. This cognitive conflict — 'my tool is not working' — creates genuine motivation for the new technique. The Think-Pair-Share surfaces misconceptions (e.g., trying $8 \times -6 = -48$ but $8 + (-6) = 2$, not 6) and lets the teacher use real learner errors as teaching material without embarrassment.	Observable indicator: Learner has correctly written $x^2 + 6x - 48 = 0$ (or equivalent $x^2 + 6x = 48$) and has recorded at least one attempted factor pair, with a note that no integer pair works. Teacher scans 5–6 books during circulation; if fewer than half have the correct equation, pause and rebuild the area-model equation setup before proceeding.
Observe Phase 📹 **VIDEO:** Comparing with multiplication: magic Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Square Roots	Each learner receives a 10 × 10 grid paper sheet (or uses the ARES digital grid tool). The teacher guides learners through a geometric construction in three steps: Step 1 — Draw an x-by-x square (labelled 'x^2'). Step 2 — Attach a 6-by-x rectangle along one side (labelled '6x'), then split it into two 3-by-x strips and place one strip on each side of the x^2 square. Step 3 — Notice the missing corner: a $3 \times 3 = 9$ square.	Distribute grid paper and say: 'We are going to build the algebra with our eyes, not just our pencils.' Model Step 1 on the board (draw x^2 square). Say: 'Draw your own x-by-x square now. Label it x^2.' WAIT 30 seconds for all learners to draw. Move to Step 2: 'I have a rectangle that is 6 wide and x tall — representing the 6x term. I will split it into two equal strips of width 3.' Draw both strips on the board. Say: 'Add yours now.' WAIT 30	Geometric Representation → Algebraic Abstraction (Concrete-Representational-Abstract, CRA): Learners construct the geometric model before writing a single algebraic step, ensuring the abstract manipulation has a visual referent. Annotating arrows from diagram regions to algebraic terms builds the dual-coding connection that prevents rote memorisation. This mirrors how Kenyan construction planners visualise	Observable indicator: Learner's grid-paper diagram shows the x^2 square, two 3×x strips correctly placed on adjacent sides, and the 3×3 corner shaded and labelled '9', with the algebraic identity $(x + 3)^2 = x^2 + 6x + 9$ written below the diagram. Teacher circulates and stamps/ticks 6 correct diagrams; for learners who have the strips in wrong positions, ask: 'Which two sides of the x^2 square are we attaching strips to?' rather than

and Irrational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners physically shade and label the missing corner, then write the algebraic identity they have just drawn: $(x + 3)^2 = x^2 + 6x + 9$. Learners then rewrite $x^2 + 6x = 48$ as $(x + 3)^2 - 9 = 48$, and finally $(x + 3)^2 = 57$. They record observations: 'What shape did we complete? What number filled the corner? Where did the 3 come from?' Learners annotate their diagrams with arrows linking geometric regions to algebraic terms.	seconds. Step 3: 'Look at the corner that is missing. What are its dimensions?' Cold-call 2 learners — WAIT 10 seconds each. Confirm: '3 by 3 — a square of area 9. Write 9 in that corner.' Say: 'We have literally COMPLETED the square. Now write in algebra what you drew.' Guide: '$(x + 3)^2 = x^2 + 6x + 9$.' Then say: 'Our equation was $x^2 + 6x = 48$. Add 9 to both sides. What do you get?' Cold-call 1 learner — WAIT 12 seconds. Write $(x + 3)^2 = 57$ on the board. Say: 'Every line of algebra you just wrote corresponds to a region on your diagram. Circle the matching pairs.' Allow 2 minutes for annotation.	floor plans before calculating areas — a context already familiar from the Githurai hall phenomenon.	correcting directly.
Explain Phase 📹 **VIDEO:** Comparing with multiplication: magic Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Square Roots and Irrational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners now generalise the geometric steps into a four-step written algorithm in their notes: Step 1 — Rearrange so $x^2 + bx = c$. Step 2 — Halve b, square it $(b/2)^2$, add to both sides. Step 3 — Write the left side as $(x + b/2)^2$. Step 4 — Take the square root of both sides and solve for x (remembering ±). They apply the algorithm to solve $(x + 3)^2 = 57$ fully: $x + 3 = \pm\sqrt{57}$, so $x = -3 + \sqrt{57} \approx 4.55$ m or $x = -3 - \sqrt{57}$ (rejected since width cannot be negative). Learners then work in pairs on two further practice equations — one from a Mombasa dhow-building context and one returning directly to the Githurai hall equation $w^2 + 2w - 120 = 0$ — and verify that completing the square gives the same answer as factorisation did in Lesson 4. Pairs write a one-sentence explanation: 'Completing the square works here because…'	Say: 'Let us pull the geometry into a recipe we can always follow.' Write the four steps on the board one at a time, pausing after each for learners to copy. For Step 2, say: 'The magic number is always $(b \div 2)^2$. In our Nakuru problem, b = 6, so the magic number is $(6 \div 2)^2 = 9$. Say that rule back to your partner now.' WAIT 15 seconds for paired rehearsal. Cold-call 1 pair: 'Kamau and Atieno, say the rule.' After solving the Nakuru equation, say: 'Negative width is physically impossible — just like a negative length of the Githurai hall makes no sense. We reject $x = -3 - \sqrt{57}$.' For the Githurai hall verification, say: 'Use completing the square on $w^2 + 2w - 120 = 0$. You should get the same width you found in Lesson 4. If you do, your two methods agree — that is powerful.' Circulate and use the prompt: 'Show me Step 2 in your working — what is your magic number?' for any learner who is stuck. After pairs finish, cold-call 3 pairs to	Algorithm Generalisation + Cross-Method Verification: Learners abstract the geometric steps into a transferable four-step procedure, then stress-test it by re-solving the familiar Githurai hall equation. Matching the Lesson 4 answer confirms the algorithm's reliability and deepens understanding of mathematical consistency. The one-sentence explanation (a micro-CER — Claim, Evidence, Reasoning) builds scientific communication skills aligned with NGSS Practice 6 (Constructing Explanations).	Observable indicator: Learner's written solution to the Nakuru garden problem shows all four algorithm steps clearly labelled, a correct 'magic number' of 9, the equation $(x + 3)^2 = 57$, and a final answer of $x \approx 4.55$ m with the negative root explicitly rejected with a reason. For the Githurai hall re-verification, the learner should obtain w = 10 m, matching the Lesson 4 result. If a learner obtains a different value, the teacher prompts: 'Compare your Step 2 magic number with a partner — are you halving b before squaring?'

		present their one-sentence explanation — WAIT 10 seconds each before taking the answer.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Comparing with multiplication: magic Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Square Roots and Irrational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes a new sticky-note (or types into the ARES DQB digital board) responding to the prompt: 'What can completing the square do that factorisation cannot? How does this help the Githurai committee?' Learners post their notes under a new DQB column headed: 'Methods that ALWAYS work.' The class reads three notes aloud and agrees on a shared DQB headline statement. Learners also revisit any earlier DQB questions — such as 'What do we do when the equation won't factorise?' — and move them from the 'Questions' side to the 'Answers' side, physically demonstrating growing understanding.	Point to the DQB and say: 'Look at the question we posted in Lesson 2 — what do we do when factorisation fails? Today you answered it. Write your sticky-note now — two sentences maximum.' WAIT 3 minutes for silent individual writing. Collect 6 notes and read 3 aloud without naming the author. Ask the class: 'Which of these best captures what we discovered?' Take a show-of-hands vote. Write the winning statement on the DQB in large print: 'Completing the square ALWAYS works, even when factorisation fails.' Then say: 'Move the sticky question about non-factorisable equations from the Questions column to the Answered column — that is YOUR progress as mathematicians.' Ask the class: 'What NEW questions do you have now?' Prompt if silent: 'Is there a single formula that does completing the square for us automatically?' Accept all responses and post new questions — this seeds Lesson 6 (the Quadratic Formula).	Driving Question Board Curation (NGSS Practice 1 — Asking Questions): The DQB move from 'Questions' to 'Answered' provides a visible, affective record of intellectual growth. Generating new questions (e.g., 'Is there a formula?') is a metacognitive act that positions learners as active knowledge-builders rather than passive receivers, and naturally previews Lesson 6 without the teacher having to announce a topic.	Observable indicator: Learner's DQB sticky-note correctly distinguishes completing the square from factorisation (e.g., references equations with non-integer roots, or says 'always works') AND makes a direct link to the Githurai hall context. A note that only restates the procedure without connecting to the phenomenon indicates the learner has procedural but not contextual understanding — flag for targeted questioning in Lesson 6.
Model Building Phase 📹 **VIDEO:** Comparing with multiplication: magic Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners open their running 'Methods for Solving Quadratic Equations' model — a structured table they have built across Lessons 3–5. They add a new row for 'Completing the Square', filling in columns: (a) When to use it, (b) Key steps, (c) Worked example (Nakuru garden), (d) Limitation (requires careful arithmetic; irrational answers), (e) Connection to Githurai hall. Learners then draw a two-column comparison	Say: 'Open your methods table from Lesson 3. We are adding Row 3 today.' Display the column headings on the board and say: 'Fill in each column for completing the square. Use your notes and the Nakuru worked example. You have 5 minutes.' Circulate and check the 'When to use it' column — target answer should include 'when the equation does not factorise easily' or 'always, as a universal method.' After 5 minutes,	Cumulative Model Revision (NGSS Practice 2 — Developing and Using Models): The running methods table is a living scientific model — each lesson adds one row, making the growth of learner understanding tangible. The side-by-side comparison diagram (factorisation vs. completing the square on the same equation) reinforces that mathematical truth is consistent across methods — a key disciplinary norm. The	Observable indicator: The 'When to use it' cell in the methods table correctly identifies completing the square as applicable to all quadratic equations, not only to those with irrational roots. The comparison diagram shows correct final values (w = 10 m) from both methods, with the double-headed arrow labelled. Exit check: teacher scans 5 books and confirms the diagram is present before dismissing the class.

Square Roots and Irrational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	diagram: on the left, the Lesson 4 factorisation tree for $w^2 + 2w - 120 = 0$; on the right, the completing-the-square solution for the same equation. They draw a double-headed arrow between the final answers and label it: 'Same answer — different paths.' Finally, learners write one prediction in their books: 'I think the next method (Lesson 6) will be…', priming curiosity about the quadratic formula.	cold-call 2 learners to read their 'Limitation' entry — WAIT 10 seconds each. For the comparison diagram, say: 'I want you to show that Lessons 4 and 5 are two roads to the same destination. Draw both solutions side by side for the Githurai hall equation.' Allow 4 minutes. Close the lesson by saying: 'The committee in Githurai now has TWO reliable tools. Next lesson, they will get a third — one formula that does everything. Write your prediction now.' WAIT 2 minutes for silent prediction writing. Collect 3 predictions to read at the start of Lesson 6.	prediction task activates anticipatory thinking, boosting retention of the upcoming quadratic formula.	

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your ARES teacher journal:

1. GEOMETRIC UNDERSTANDING: Could learners correctly link the 3 × 3 corner square in their diagram to the algebraic step of adding $(b/2)^2$ to both sides, or did they treat the geometry and algebra as separate activities? If the link was weak, plan a brief 5-minute diagram-to-algebra matching warm-up at the start of Lesson 6.

2. THE MAGIC NUMBER: How many learners consistently applied $(b/2)^2$ correctly vs. using b/2 without squaring, or squaring b before halving? Tally from your clipboard. If more than one-third of the class made this error, design a targeted 3-question drill (30-second individual whiteboards) at the start of Lesson 6 before introducing the quadratic formula.

3. CONNECTING TO PHENOMENON: When learners re-solved the Githurai hall equation using completing the square in Phase 3, did they spontaneously connect back to the committee's planning problem, or did they treat it as an abstract exercise? If connection was low, increase the frequency of explicit phenomenon callbacks in Lessons 6 and 7 — e.g., start every worked example with 'The Githurai committee now needs to solve…'

4. DQB QUALITY: Were the new questions posted on the DQB genuinely inquiry-oriented (e.g., 'Is there a single formula?', 'What if b is odd?') or mostly procedural ('What is the magic number?')? Inquiry-level questions indicate readiness for the quadratic formula derivation in Lesson 6; procedural questions suggest learners need more consolidation time — consider assigning the Nakuru garden problem as homework with a reflection prompt.

5. PACING: The geometric construction in Phase 2 tends to run long if learners are drawing freehand. Note whether you needed to shorten Phase 5. If so, the model-table update can be completed as a structured homework task with a short peer-verification at the start of Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that the Nakuru school garden equation ($x^2 + 6x = 48$) could not be solved by factorisation over integers. Through a geometric grid-paper construction, they saw a literal square being 'completed' by adding a 3 × 3 corner piece, and observed that the resulting diagram corresponded exactly to the algebraic identity $(x + 3)^2 = x^2 + 6x + 9$.
What did I learn?	Learners learned that any quadratic equation of the form $x^2 + bx = c$ can be rewritten as $(x + b/2)^2 = c + (b/2)^2$ — a process called completing the square. They learned that this method always works, produces exact (sometimes irrational) solutions, and gives the

	same answer as factorisation when both methods apply. They applied this to the Nakuru garden and re-verified the Githurai hall dimensions.
How does this explain the phenomenon?	Learners explained why completing the square works by connecting each algebraic step to a region in their geometric diagram. They explained that the Githurai committee can now use this method for any rectangular-area planning problem — even when the dimensions do not produce neat integer factor pairs — and updated the DQB with the headline: 'Completing the square ALWAYS works, even when factorisation fails.'

LESSON 6 (80 minutes): The Quadratic Formula & The Mombasa Stone Problem	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students derive the quadratic formula by completing the square on the general form $ax^2 + bx + c = 0$, apply it to real Kenyan contexts (the Mombasa projectile problem and the mandazi pricing problem), interpret the discriminant to determine the number of real solutions, and update the Driving Question Board with the understanding that the quadratic formula is a universal solving tool.
Knowledge	By the end of this lesson, the learner should be able to: (a) derive the quadratic formula by completing the square on $ax^2 + bx + c = 0$; (b) state the quadratic formula $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$; (c) define the discriminant $\Delta = b^2 - 4ac$ and interpret its value ($\Delta > 0$: two real solutions, $\Delta = 0$: one real solution, $\Delta < 0$: no real solutions); (d) identify the values of a, b, and c from any quadratic equation written in standard form; (e) apply the quadratic formula to solve equations arising from real-world contexts.
Skills	By the end of this lesson, the learner should be able to: (a) correctly derive the quadratic formula through a guided algebraic sequence; (b) substitute values of a, b, and c into the quadratic formula accurately; (c) use a scientific calculator to evaluate square roots and verify solutions; (d) compare solutions obtained by the quadratic formula with those obtained by factorisation and completing the square in earlier lessons; (e) solve applied problems modelled by quadratic equations using the formula.
Attitudes	By the end of this lesson, the learner should be able to: (a) appreciate that a single universal formula can solve any quadratic equation, reflecting the power of mathematical generalisation; (b) show persistence when performing multi-step algebraic derivations; (c) value the cross-checking of solutions using different methods as a form of intellectual honesty; (d) recognise mathematics as a tool for addressing real community planning challenges such as the Githurai Harambee hall.
Key Inquiry Question	How can we use quadratic equations to model and solve problems in day-to-day activities?
Purpose in Storyline	In Lessons 1–5, students built up a toolkit: they identified quadratic structures, expanded and factorised expressions, solved by factorisation, and solved by completing the square. Each method worked — but each also had limitations (factorisation requires 'nice' factors; completing the square is lengthy). Lesson 6 answers a natural question that has been growing on the DQB: 'Is there ONE method that always works?' Students derive the quadratic formula — essentially completing the square once, on the general form — and immediately apply it to the Mombasa stone projectile problem ($-5t^2 + 20t + 25 = 0$) and the mandazi pricing problem. They also meet the discriminant as a quick screening tool. This positions Lesson 7 (the capstone) where students will choose the best method to finally resolve the Githurai Harambee hall dimensions and present a full solution to the community.
Safety Notes	No physical hazards in this lesson. Ensure all students handle scientific calculators responsibly. Remind learners that calculator batteries can fail — manual verification of formula steps is always required. If the lesson is delivered in a computer lab or using tablets on the ARES platform, students should save work frequently.

B. LESSON OVERVIEW
Lesson 6 of 7 in the quadratic equations evidence-gathering sequence. Students work in groups of 4–5. The lesson opens by revisiting the Githurai Harambee hall phenomenon and the DQB question 'Is there a method that works for every quadratic?' Students are guided through a structured derivation of the quadratic formula by completing the square on $ax^2 + bx + c = 0$, recording each step on mini-whiteboards. They then apply the formula to two Kenya-context problems: (1) the Mombasa projectile problem — a stone thrown upward from a cliff modelled by $h(t) = -5t^2 + 20t + 25$, asking when the stone hits the sea ($h = 0$); and (2) the mandazi pricing problem from earlier lessons. Students use scientific calculators to verify solutions, then explore the discriminant by testing three different equations and classifying the number of real solutions. The lesson closes with a DQB update and a model revision connecting the formula back to the Githurai hall problem.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students are shown three quadratic equations on the board: (A) $x^2 - 5x + 6 = 0$, (B) $2x^2 - 4x + 2 = 0$, and (C) $x^2 + x + 3 = 0$. They are asked: 'Which of these equations do you think has two solutions? One solution? No real solutions? Write your prediction and a one-sentence reason in your exercise books. Do NOT solve yet.' Students write individually for 3 minutes, then turn and share with their elbow partner. The teacher takes a quick class poll by show of hands for each equation, recording tallies on the board. Most students will likely predict incorrectly for (C), surfacing the need for a reliable screening tool — the discriminant.	1. Display the three equations on the board (or ARES screen) and read them aloud. Say: 'Before we learn anything new today, I want to know what your mathematical instincts tell you. Which equation has two answers? Which has one? Which has none?' 2. Allow 3 minutes of silent individual writing — enforce silence with a visible countdown timer. 3. Say: 'Now turn to your elbow partner — you have 90 seconds. Compare predictions and try to agree.' 4. After partner talk, say: 'Hands up — who thinks equation A has two solutions? [Tally.] Who thinks B? Who thinks C?' Record tallies next to each equation. Do NOT confirm or deny. 5. Cold-call 3 students (one per equation) and ask: 'Explain your reasoning in one sentence.' WAIT 12 seconds after each question before accepting answers. 6. Say: 'Keep your predictions visible — by the end of today you will know exactly how to check this without solving the full equation. That tool is called the discriminant, and it comes from something even bigger: the quadratic formula.'	Prediction with Justification — students are required to give a reason, not just a guess. This activates prior knowledge (number of solutions relates to the graph, which some students may recall), surfaces misconceptions (e.g., 'if the numbers are big it has no solution'), and creates cognitive dissonance that motivates the derivation. The class tally makes disagreement visible and productive.	Observable indicators: (a) Students write a prediction AND a reason — teacher circulates and flags students who write only a guess with no reasoning (these students need prompting: 'What do you remember about the graph of a quadratic?'). (b) The class tally reveals whether students have any prior intuition about the discriminant. If more than 60% predict correctly for all three, the teacher can accelerate Phase 2; if fewer than 40% are correct, explicitly say 'Most of us are guessing — that is exactly why we need the formula and the discriminant today.'
Observe Phase 📹 **VIDEO:** Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum) 🔍 Search ARES	PART A — Deriving the Formula (25 minutes): Each group receives a printed 'Derivation Scaffold' card listing the steps of completing the square on $ax^2 + bx + c = 0$, but with key algebraic expressions blanked out. Groups work through the scaffold together, filling in each blank step-by-step on mini-whiteboards. Steps scaffolded: (1)	PART A: 1. Distribute Derivation Scaffold cards. Say: 'This card shows you the skeleton of the most important derivation in this whole unit. Your job as a group is to fill in the missing algebra — you already know every single step because you did completing the square in Lesson 5. This is just doing it with letters instead of	Structured Derivation with Visible Verification (mini-whiteboards) — by filling in a scaffold rather than copying notes, students actively construct the formula rather than passively receive it. Holding up boards at each step creates a whole-class checkpoint and allows the teacher to identify errors in real time before they compound. The	Observable indicators: (a) At each whiteboard check-in during the derivation, teacher looks for: correct handling of the $(b/2a)^2$ term (common error: students write $b^2/2a$ instead of $b^2/4a^2$); correct ± sign placement; correct final formula. (b) In the Mombasa problem, the key observable is whether students reject $t = -1$ with

for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Write $ax^2 + bx + c = 0$. (2) Divide by a: $x^2 + (b/a)x + c/a = 0$. (3) Move constant: $x^2 + (b/a)x = -c/a$. (4) Complete the square: $x^2 + (b/a)x + (b/2a)^2 = -c/a + (b/2a)^2$. (5) Write as perfect square: $(x + b/2a)^2 = (b^2 - 4ac)/4a^2$. (6) Square root both sides: $x + b/2a = \pm\sqrt{(b^2 - 4ac)}/2a$. (7) Isolate x: $x = (-b \pm \sqrt{(b^2 - 4ac)})/2a$. Groups hold up mini-whiteboards at each step for teacher verification before proceeding. PART B — Mombasa Stone Problem (15 minutes): Students read a scenario card: 'During a Harambee celebration in Mombasa's Old Town, a stone is thrown upward from the top of a 25-metre cliff near the sea. Its height above the water is given by $h(t) = -5t^2 + 20t + 25$, where t is in seconds and h is in metres. When does the stone hit the water?' Students set h = 0, write $-5t^2 + 20t + 25 = 0$, identify a = −5, b = 20, c = 25, substitute into the quadratic formula, simplify using calculators, and interpret both solutions (t = −1 is rejected since time cannot be negative; t = 5 seconds is the answer). PART C — Mandazi Pricing Problem (8 minutes): Students revisit the mandazi problem from Lesson 4: a vendor's daily profit is modelled by $P = -2x^2 + 40x - 150$, where x is price per mandazi in shillings. They find the break-even prices (P = 0) using the quadratic formula and verify using their calculator.	numbers.' 2. Circulate and monitor group progress. At Step 4 (completing the square term), anticipate the hardest step: pause the class and say: 'Everyone stop — what is half of b/a? [WAIT 12 seconds.] Cold-call: [Name 1], what did your group write?' Then cold-call a second group: '[Name 2], do you agree? Why?' 3. At Step 6, emphasise: 'Notice the ± sign — this is where both solutions come from. Write that in a box in your notes.' 4. When groups finish, say: 'Hold up your whiteboard showing the final formula. Look around — does every board show the same thing?' Confirm and write the boxed formula on the board: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$. PART B: 5. Read the Mombasa stone scenario aloud with expression. Say: 'Wapi mwanafizikia wa Mombasa? — Where is our Mombasa physicist?' (light tone to build engagement). 6. Say: 'Step one — always write the equation in standard form. What is a? What is b? What is c? Write them down before substituting. [WAIT 10 seconds.] [Name 3], tell me your values.' 7. After groups calculate, say: 'You should have two values of t. Are both physically meaningful? Discuss with your group for 60 seconds.' Cold-call: '[Name 4], which solution do you reject and why?' WAIT 12 seconds. 8. Say: 'This is exactly the kind of thinking the Githurai committee needs — equations give you mathematical answers, but YOU must interpret what makes sense in context.' PART C: 9. Say: 'You have 8 minutes — apply the formula to the mandazi problem. Use your calculator. Compare your answers to what you found in Lesson 4 using	two applied problems immediately contextualise the abstract formula in Kenyan settings, reinforcing the lesson's storyline thread.	a valid reason — if they accept both solutions, teacher must address the 'context-filtering' skill. (c) In the mandazi problem, students should obtain x = 5 and x = 15 — if calculator answers differ, teacher checks sign errors in identifying a, b, c. (d) Circulate and note which groups are still struggling with identifying a, b, c in non-standard forms — flag these students for targeted support in Phase 3.

		factorisation.' Circulate, checking that students are correctly identifying a = −2, b = 40, c = −150 and not making sign errors.		
Explain Phase 📹 **VIDEO:** Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	PART A — Discriminant Investigation (10 minutes): Students return to the three equations from Phase 1 (Predict). For each equation, they calculate $b^2 - 4ac$ without solving fully, record the value, and classify it (positive, zero, or negative). They then solve all three fully using the quadratic formula and observe: equation A gives two distinct solutions; equation B gives one repeated solution; equation C involves √(negative number) and they see it has no real solutions. Students complete a summary table in their books: Equation | a | b | c | $\Delta = b^2-4ac$ | Nature of solutions. PART B — Method Comparison Table (8 minutes): Groups complete a comparison table in their exercise books contrasting factorisation, completing the square, and the quadratic formula across four criteria: (1) speed, (2) always works?, (3) gives exact vs decimal answers, (4) best used when? Students discuss: 'Which method would you use for the Githurai hall problem ($w^2 + 2w - 120 = 0$)? Can you factorise it? Can the formula handle it?' They confirm that $w^2 + 2w - 120 = 0$ CAN be factorised (tested in earlier lessons) but the formula would also work.	PART A: 1. Say: 'Go back to your Phase 1 predictions. Now calculate $b^2 - 4ac$ for each equation — just that expression, nothing more yet. [WAIT 15 seconds for students to start.] 2. After 3 minutes, cold-call three different students — one per equation — to state their discriminant value. WAIT 12 seconds each time. Record values on the board next to the tallies from Phase 1. 3. Say: 'Now solve equation C using the formula. What happens when you try to find the square root of a negative number on your calculator? [WAIT 10 seconds.] [Name 5], what did your calculator display?' Expected: ERROR or a non-real result. 4. Say: 'That error message is mathematics telling you something important. Equation C has no real solutions — and you could have known that in five seconds just by checking the discriminant. This is the power of Δ.' Write on the board: $\Delta > 0 \rightarrow$ 2 real solutions | $\Delta = 0 \rightarrow$ 1 real solution | $\Delta < 0 \rightarrow$ no real solutions. 5. Return to Phase 1 tallies: 'How many of us predicted correctly? Were our instincts right?' Celebrate correct predictions and acknowledge the challenge. PART B: 6. Say: 'You have now used three methods to solve quadratic equations. Take 8 minutes to complete the comparison table in your exercise books. Be honest — when is factorisation faster? When is the formula safer?' 7. After 5 minutes, ask: 'For the Githurai hall problem	Claim-Evidence-Reasoning (CER) applied to method selection — students are not just practising procedures; they are making claims ('factorisation is faster'), supporting them with evidence (the comparison table), and reasoning about when each tool is appropriate. This builds metacognitive awareness of mathematical methods, which is a core CBE competency. The return to Phase 1 predictions closes the cognitive loop opened at the start of the lesson.	Observable indicators: (a) Discriminant table is complete and correct — teacher checks that students have correctly computed Δ for all three equations (A: $\Delta = 1$, B: $\Delta = 0$, C: $\Delta = -11$) and matched the correct nature of solutions. (b) Method comparison table shows nuanced reasoning, not just 'formula is best' — students should identify that factorisation is faster for 'nice' equations. (c) When asked about the Githurai hall equation, students should be able to confirm it is factorisable $(w + 12)(w - 10) = 0$ — this is a direct preview of Lesson 7. Students who cannot recall this from earlier lessons need a quick prompt back to their Lesson 3 notes.

		— $w^2 + 2w - 120 = 0$ — which method is fastest?' Cold-call [Name 6] and [Name 7] for potentially different answers. WAIT 12 seconds each. Validate both if correct: 'Both work — factorisation is faster here because the factors are integers; the formula is more powerful for non-factorisable equations.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each student receives one sticky note (or enters a response on the ARES platform if using digital devices). They respond to the DQB prompt: 'What new understanding do you have about solving quadratic equations — and how does today's lesson change what you would do to solve the Githurai hall problem?' Students write their response, then post it under the relevant DQB column: 'What we now know' or 'What we still wonder'. The class reads 4–5 posted notes aloud. The teacher writes the lesson's anchor statement on the DQB: 'THE QUADRATIC FORMULA is a universal tool — it solves any quadratic equation. The discriminant ($b^2 - 4ac$) tells us how many real solutions exist before we fully solve.' Students also update the 'Methods Available' section of the DQB to now show three methods: Factorisation | Completing the Square | Quadratic Formula.	1. Say: 'We started this unit with a committee in Githurai who couldn't find the hall dimensions. Look at our DQB — we have been building a toolkit lesson by lesson. Today we added the most powerful tool: the quadratic formula. Take 3 minutes to write your sticky note response.' 2. Circulate and read notes as students write. Identify 2 'What we still wonder' notes that raise genuine mathematical questions (e.g., 'What if a = 0?' or 'Can you get a complex number as an answer?') — these will be used to close the lesson. 3. After posting, read 3 'What we now know' notes and 2 'What we still wonder' notes aloud to the class. For each 'wonder' note, say: 'That is a great mathematical question. Write it in your exercise book — it may be explored in future mathematics courses.' 4. Write the anchor statement on the DQB in a new colour (e.g., green) to mark it as Lesson 6's contribution. 5. Point to the Githurai hall problem on the DQB: '$w^2 + 2w - 120 = 0$. We could now solve this with any of our three methods. In Lesson 7 — our final lesson — you will choose your method, solve it completely, interpret the answer, and present your solution as if you were reporting back to the Githurai Harambee committee.'	Collaborative DQB Updating — the public, cumulative nature of the DQB makes students' growing understanding visible over time. By physically adding to the board and seeing it fill up across lessons, students experience a concrete narrative of intellectual progress. Connecting today's entry to Lesson 7's culminating task creates forward momentum and purpose.	Observable indicators: (a) Sticky notes reference the quadratic formula, the discriminant, or method comparison — generic notes ('I learned about quadratic equations') indicate surface-level engagement and require a teacher follow-up question: 'Can you be more specific about what the formula tells us?' (b) The 'What we still wonder' notes reveal whether students are thinking beyond procedures (e.g., wondering about complex numbers or the graphical meaning of the discriminant) — these are signs of higher-order engagement. (c) Students who cannot connect today's learning to the Githurai hall problem need a direct prompt: 'What are the values of a, b, and c for the Githurai equation?'

Model Building Phase 📹 **VIDEO:** [Using the quadratic formula: number of solutions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative 'Quadratic Equations Toolkit' model — a visual diagram they have been building since Lesson 1. Today they add a new layer: the Quadratic Formula branch. On their model, they draw a new branch labelled 'Method 3: Quadratic Formula', adding the formula $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$, the discriminant classification table ($\Delta > 0$, $= 0$, < 0), and an annotated example using the Mombasa stone problem ($-5t^2 + 20t + 25 = 0 \rightarrow t = 5$ s). They also draw a connecting arrow from 'Method 3' back to the Githurai hall problem ($w^2 + 2w - 120 = 0$) with a note: 'Formula works here — but factorisation is faster.' Finally, students write one sentence at the bottom of their model: 'The best method depends on the equation — a mathematician chooses tools wisely.'	1. Say: 'Take out your Toolkit model from Lesson 1. You are about to add the final method branch before our big Lesson 7 presentation. You have 8 minutes — work individually, but you may ask your group for help on the diagram layout.' 2. Circulate and check that: (a) the formula is written correctly (common error: missing the 2a in the denominator — the whole expression $-b \pm \sqrt{b^2-4ac}$ must be divided by 2a, not just $\pm\sqrt{b^2-4ac}$); (b) the discriminant table shows all three cases; (c) the Mombasa problem is used as the worked example, not a different problem. 3. If a student has written the formula incorrectly, do NOT correct it for them — say: 'Check your formula against Step 7 of your Derivation Scaffold card. Do they match?' WAIT for the student to self-correct. 4. With 2 minutes remaining, say: 'Look at your whole model — from Lesson 1 to today. You have built a complete toolkit for solving quadratic equations. Tomorrow in Lesson 7, you use this toolkit to finally answer the committee's question and solve the Githurai hall problem completely.' 5. Collect or photograph all models for teacher review before Lesson 7.	Cumulative Model Building — the ongoing Toolkit model serves as a personalised knowledge representation that grows with each lesson. Adding to an existing model (rather than starting fresh) reinforces that knowledge is cumulative and connected. The explicit comparison arrow ('works here — but factorisation is faster') develops metacognitive tool-selection skills, a core mathematical practice aligned to NGSS Science and Engineering Practice 5 (Using Mathematics and Computational Thinking).	Observable indicators: (a) The formula on the student's model matches the board-written formula exactly — any discrepancy (especially in the denominator or the ± placement) must be corrected before Lesson 7. (b) The discriminant table shows all three cases with correct inequality/equality signs. (c) The Mombasa example is worked correctly on the model ($t = 5$ s, with $t = -1$ rejected). (d) The connecting arrow to the Githurai equation is present — students who omit this have not yet connected the new learning to the driving question and need a direct prompt. (e) Teacher photographs all models and reviews them before Lesson 7 to identify students who will need targeted support during the Lesson 7 capstone activity.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. DERIVATION ACCESSIBILITY: Did all groups successfully complete the Derivation Scaffold, or did certain steps (especially the $b^2/4a^2$ completing-the-square term) cause widespread confusion? If more than 30% of groups struggled at Step 4, consider opening Lesson 7 with a 5-minute 'formula verification' warm-up where students substitute the Githurai values $a=1$, $b=2$, $c=-120$ into the formula and confirm they get the same answers as in Lesson 3 (factorisation).

2. CALCULATOR FLUENCY: Did students confidently use scientific calculators to evaluate $\sqrt{b^2-4ac}$? If students were slow or made calculator errors (especially with negative values under the square root), plan to address calculator technique at the start of Lesson 7.

3. CONTEXT REJECTION: When solving the Mombasa stone problem, did students spontaneously reject $t = -1$ as physically meaningless, or did they need significant

prompting? Students who accepted both solutions without questioning real-world meaning may struggle with the Lesson 7 capstone interpretation ($w = -12$ must be rejected since width cannot be negative).

4. DQB QUALITY: Were the sticky notes specific and connected to the driving question? If notes were generic, consider assigning a more structured prompt for Lesson 7's DQB update: 'Complete this sentence: The Githurai committee should use [METHOD] because ___.'

5. PACING: The lesson is content-heavy (derivation + two applied problems + discriminant + model building). If time ran short, note which phase was compressed and ensure that phase is revisited in Lesson 7's opening. The derivation (Phase 2 Part A) is the highest-priority activity and should not be rushed — it is the conceptual centrepiece of the entire lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed the algebraic derivation of the quadratic formula by completing the square on the general form $ax^2 + bx + c = 0$, step by step using a Derivation Scaffold card. They also observed two applied Kenyan scenarios: (1) a stone thrown from a Mombasa cliff modelled by $h(t) = -5t^2 + 20t + 25 = 0$, and (2) a mandazi vendor's break-even pricing modelled by $-2x^2 + 40x - 150 = 0$. Students observed that one of the Mombasa solutions ($t = -1$) had to be rejected because time cannot be negative.
What did I learn?	Students learned that: (1) the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$ can be derived by completing the square on any quadratic equation in standard form; (2) the discriminant $\Delta = b^2 - 4ac$ determines the number of real solutions before full solving ($\Delta > 0$ gives two solutions, $\Delta = 0$ gives one, $\Delta < 0$ gives none); (3) the quadratic formula works for every quadratic equation, even those that cannot be factorised easily; (4) mathematical solutions must always be interpreted in context — not all algebraic solutions are physically or logically meaningful.
How does this explain the phenomenon?	Students explained, using the Claim-Evidence-Reasoning strategy and their method comparison table, when each of the three methods (factorisation, completing the square, quadratic formula) is most appropriate. They explained why $t = -1$ is rejected in the Mombasa problem and why the quadratic formula is described as a 'universal tool'. They also explained, using the discriminant, why equation C ($x^2 + x + 3 = 0$) from the Predict phase has no real solutions, revisiting and correcting their initial predictions.

LESSON 7 (40 minutes): Application, Model Building and Consolidation — Closing the Harambee Hall Story

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of this lesson, students will demonstrate mastery of all three methods for solving quadratic equations and apply them to authentic Kenyan community problems, justifying their chosen approach and communicating mathematical reasoning clearly.
Knowledge	Students will know: (1) the three solution methods — factorisation, completing the square, and the quadratic formula — and the conditions that make each most efficient; (2) how the discriminant predicts the nature of solutions before full solving; (3) how to interpret both solutions of a quadratic equation in context and reject non-physical answers; (4) how the full learning journey from expanding expressions to solving equations forms a coherent mathematical toolkit.
Skills	Students will be able to: (1) solve quadratic equations using the method best suited to each problem; (2) model new community-based scenarios as quadratic equations in standard form; (3) verify solutions by substitution and contextual reasoning; (4) construct and present a cumulative visual model that maps the mathematical journey of the unit; (5) communicate mathematical justifications clearly in spoken and written form.
Attitudes	Students will appreciate that mathematics is a practical tool for community empowerment; show confidence in selecting and defending a method; demonstrate collaborative responsibility in group presentations; and value the iterative nature of mathematical sense-making as reflected in the Driving Question Board journey.
Key Inquiry Question	How can we use quadratic equations to model and solve problems in day-to-day activities — and which solution method should a community problem-solver choose?
Purpose in Storyline	This final lesson closes the narrative arc that began in Lesson 1 when the Githurai Harambee committee could not find the hall dimensions. Students now return to that original problem with a full toolkit, present complete solutions by all three methods, close every card on the Driving Question Board, and synthesise the unit into a community-facing model — demonstrating that mathematical learning has genuine social value.
Safety Notes	No physical hazards. Ensure group work is inclusive — assign roles so no student is sidelined. Validate diverse presentation styles including oral, visual, and written. Emotional safety: affirm that all three methods are valid and that choosing one over another is a matter of mathematical judgement, not correctness.

B. LESSON OVERVIEW

In this 40-minute consolidation lesson, students return triumphantly to the Githurai Harambee hall scenario and solve it completely using factorisation, completing the square, and the quadratic formula, comparing efficiency and justifying their preferred method. Groups then tackle new Kenyan-context problems — a Kisumu fish-drying yard, a Nakuru school fundraising table, and a Githurai construction border — modelling each as a quadratic equation and solving it. In the Model Building phase, groups create a cumulative journey poster or slide showing the progression from quadratic expressions to community problem-solving. The Driving Question Board is formally closed as every initial wonder question is answered with evidence from the lesson sequence. The lesson ends with individual written reflection connecting quadratic thinking to community empowerment.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Using the quadratic formula: number of solutions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students enter to find the original manila-paper sketch from Lesson 1 — the rectangle labelled width = w, length = w + 2, Area = 120 m² — displayed prominently at the front, alongside their own Lesson 1 prediction cards. The teacher asks students to silently read their own original predictions. Students then write on a half-sheet: (a) What was my Lesson 1 prediction for the width? (b) What do I now know the width is, and how do I know? (c) Which of the three methods would I choose to solve this problem today, and why? Students share their half-sheets with a shoulder partner for 90 seconds before a whole-class show of hands: factorisation, completing the square, or quadratic formula. The spread of choices is noted visibly on the board to motivate the comparative activity that follows.	Teacher holds up a Lesson 1 prediction card (one that shows a wrong guess) and says warmly: 'Six weeks ago, someone in this room wrote 9 as their answer, arrived at by trial and error. Today that same student — and all of you — can prove the answer using three different methods of algebra. That is the journey we are closing today.' Teacher then poses the orienting question: 'If you were advising the Githurai committee right now, which method would you recommend they use to find the hall dimensions — and could you explain why to a person who has never studied quadratics?' WAIT TIME: Allow 30 seconds of silent individual thinking before the partner share. After the show of hands, say: 'We have three camps in this room. By the end of today, each camp will present their method, and the class will decide together which is most practical for a community setting.' Record the tally — e.g. Factorisation: 14, Completing the Square: 6, Formula: 8 — on the board and leave it visible throughout.	Activating prior knowledge through self-assessment against original predictions; anchoring the lesson goal in a genuine community decision (which method to recommend); creating productive disagreement among method camps to motivate comparative analysis.	Teacher reads the half-sheet responses during the partner-share, circulating to identify: (a) students who can correctly state w = 10 and explain why w = -12 is rejected; (b) students who can articulate a reasoned method preference; (c) students who still conflate the three methods. These observations guide grouping in the Observe phase.
Observe Phase 📹 **VIDEO:** Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear	Students are organised into three expert groups — Group F (Factorisation), Group C (Completing the Square), and Group Q (Quadratic Formula) — each assigned to solve the Harambee hall equation w² + 2w − 120 = 0 using only their assigned method, written on A3 paper with full working. Each group also answers: (1) What are the two algebraic solutions? (2) Which solution is valid for the hall, and why? (3) What is one advantage of	During group work, circulate with targeted questions: to Group F — 'Which factor pair of 120 gives a difference of 2? How does that connect to your b value?'; to Group C — 'When you write (w + 1)² = 121, what does taking the square root of 121 tell you about the two solutions?'; to Group Q — 'Before you substitute into the formula, what does the discriminant tell you about how many real solutions to expect?' WAIT TIME: After each group presents, pause for 20	Jigsaw expert-group presentation so all three methods are observed and compared in a single episode; public verification by substitution to model mathematical discipline; generative questioning to show that solving one problem opens further mathematical thinking.	Teacher photographs each A3 sheet for later review. During presentations, teacher listens for: correct application of Zero Product Property (Group F), correct completion of the square with the constant term moved first (Group C), correct substitution of a = 1, b = 2, c = -120 into the formula (Group Q). Any procedural error is flagged with 'I notice a step here — can the presenting group talk me through it?' rather than direct correction, preserving group

Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	your method for a community member with limited calculator access? After 8 minutes of group work, each group presents their solution in under 90 seconds. The class verifies that all three methods give w = 10 and w = -12, and that w = 10 metres is the valid physical width, making the hall 10 m wide and 12 m long. Students then individually verify by substitution: $10 \times 12 = 120$ m². A brief class discussion follows: 'The committee now knows the hall is 10 m by 12 m. What new community questions does this open?' Students generate two or three follow-up questions — e.g. how much roofing material is needed, what is the perimeter for walling — which are noted on the DQB as real-world extensions.	seconds and ask the class: 'Does the working on this A3 sheet match what your group got? If not, where does it diverge?' During the verification step, say: 'Mathematicians always check their answer. Substitute w = 10 into the original equation — do not assume correctness just because the algebra looks neat.' After the follow-up question generation, affirm: 'Every question you just raised is a new quadratic or geometric problem the committee will face. You have given them a toolkit, not just one answer.'		ownership.
Explain Phase 📹 **VIDEO:** Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students now work in new mixed groups (one member from each expert group), each tackling one of three new Kenyan-context problems printed on cards. Problem A — Kisumu Fish-Drying Yard: A fishing cooperative in Kisumu needs a rectangular drying yard with area 84 m². The length must be 5 m more than the width. Write and solve the quadratic equation to find the dimensions. Problem B — Githurai Construction Border: The Harambee committee wants to add a uniform tiled border of width x around the 10 m by 12 m hall floor, so the total floor-plus-border area is 180 m². Write and solve the quadratic equation to find x. Problem C — Nakuru School Fundraising: A school in Nakuru sells chapati at a price of p shillings each. Revenue $R = p(80 - p)$. Find the price p that gives	Introduce the three problems by saying: 'These are not invented problems — fish drying, tiled borders, and food pricing are real planning challenges in Kenyan communities right now. Your job is to show that quadratic equations are practical tools, not abstract puzzles.' During group work, use targeted prompts: for Problem A — 'Before you expand, write $w(w + 5) = 84$. What is the standard form? What is a, b, c?'; for Problem B — 'The new total dimensions are (10 + 2x) by (12 + 2x). Expand that product carefully.'; for Problem C — '$p(80 - p) = 1500$ — rearrange this before choosing your method. What does the discriminant tell you?' WAIT TIME: At the gallery walk transition, say: 'You have 60 seconds at each station. Read the full working before writing anything.' After the gallery walk, bring the class together: 'Did any	Authentic new-context problems require students to transfer, not just repeat; gallery walk creates peer review and accountability without teacher arbitration; returning to the Predict-phase tally creates a visible narrative of conceptual growth within a single lesson.	Teacher collects A3 sheets and yellow sticky-note questions after the gallery walk. Formative indicators: (a) correct construction of the quadratic in standard form from a word problem (modelling competency); (b) appropriate method selection with justification; (c) correct contextual interpretation of both roots. Yellow sticky notes with mathematical questions (not just 'I do not understand') signal deeper engagement and are used to prioritise the Model Building discussion.

	revenue of 1 500 shillings. Each group must: (1) define the variable and write the quadratic in standard form; (2) choose a solution method and state why; (3) solve fully and verify; (4) interpret both solutions in context and state which is valid. Groups record work on A3 paper. After 10 minutes, groups do a gallery walk — rotating to the next problem card and A3 sheet, adding a green tick if they agree with the working or a yellow sticky note with a question if they do not.	group find a station where both algebraic solutions were physically valid? What does that mean?' Highlight that Problem C may yield two valid prices, and that context — not just algebra — determines whether we accept one or both. Close the Explain phase by returning to the board tally from the Predict phase: 'Has anyone changed their preferred method after seeing all three problems? Why?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The full DQB — all cards posted across Lessons 1 through 6 — is displayed. The teacher reads aloud each original 'What we wonder' card from Lesson 1. For each card, a volunteer identifies which lesson and which activity provided the answer, and places a green 'ANSWERED' sticker on the card. Sample wonder cards and their closures: 'Why can we not use a linear equation for area problems?' — closed in Lesson 1 and confirmed today; 'How do we expand two brackets?' — closed in Lesson 2; 'How do we reverse expansion?' — closed in Lesson 3; 'Can an equation have two correct answers?' — closed in Lesson 4 and enriched today; 'What do we do when factorisation fails?' — closed in Lessons 5 and 6; 'Is there one formula that always works?' — closed in Lesson 6. Students then write on a new card the most important thing they personally learned across the unit and post it in the 'We now know' column. Finally, if any genuine new question has emerged — e.g. 'What happens when the discriminant is negative in a real problem?' — it is posted in a new	Stand beside the DQB and facilitate with genuine celebration: 'Every single card on this board was written by someone in this room six weeks ago. You were honest enough to admit you did not know something. Now look at what you know.' Read each wonder card slowly, pausing for volunteers. If no volunteer comes forward for a card, prompt: 'Which lesson do you remember when this clicked for you?' WAIT TIME: After all cards are stickered, give 90 seconds of silent individual writing for the personal 'We now know' card before posting. When a student raises a new question for 'Beyond this unit', respond: 'That is exactly the kind of question that takes someone into Form 4 Further Mathematics or university engineering. Write it down — it belongs to you.' Close the DQB phase by returning to the Driving Question itself: 'How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems — like building the Harambee hall in Githurai?' Ask: 'Can we now answer this fully?' Accept two or	Formal DQB closure makes the arc of learning visible and celebratory; the 'Beyond this unit' column preserves intellectual honesty and models how mathematics is open-ended; connecting the final DQB moment to the original Driving Question creates narrative closure without foreclosing curiosity.	Teacher reads all 'We now know' cards as students post them, noting: (a) whether students identify conceptual learning (not just procedural steps); (b) whether any card reveals a residual misconception that needs individual follow-up after the lesson; (c) the quality of 'Beyond this unit' questions as an indicator of mathematical curiosity. Cards with procedural-only statements (e.g. 'I learned the formula') are noted for students who may benefit from additional conceptual discussion.

	'Beyond this unit' column, honouring intellectual curiosity without leaving false gaps.	three student answers verbally, then confirm: 'Yes — and you just proved it with three different methods, three different community problems, and a 10-by-12-metre hall that the Githurai committee can now actually build.'		
Model Building Phase 📹 **VIDEO:** [Using the quadratic formula: number of solutions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	In their mixed groups, students create a Cumulative Journey Model on A3 paper (or a shared digital slide if devices are available) that maps the entire unit as a connected story. The model must include: (1) The Phenomenon — a small sketch of the Githurai hall rectangle with w and w + 2 labelled; (2) The Mathematical Journey — a flow diagram showing: Recognising quadratic expressions → Expanding using FOIL → Factorising → Solving by factorisation → Completing the square → Quadratic formula; (3) The Three Methods compared in a small table showing: method name, when to use it, one example equation solved; (4) A Community Impact statement — one sentence explaining how quadratic equations help Kenyan communities make real planning decisions; (5) One new problem the group invented, set in a Kenyan context, with the equation written in standard form (solution optional — the act of modelling is the target). Groups share their models in a 60-second 'speed gallery' where each group stands by their poster and the class rotates. Students use dot stickers to vote on the model that best communicates the mathematical journey to someone who has never studied quadratics — e.g. a parent or community committee	Frame the task with purpose: 'This model is not for me or for an exam. It is for the Githurai committee, for your parents, for any Form 1 student who will study this next year. Make it clear enough that someone outside this room can follow the journey.' During construction, prompt groups who are listing steps without connection: 'Can you draw an arrow between factorisation and the Zero Product Property to show why one leads to the other?' For groups who finish early: 'Add a second community problem to your model — set it in a different county of Kenya.' During the speed gallery, circulate and listen to the 60-second pitches: prompt presenters with 'Why does the community need quadratic equations — not just arithmetic?' After the dot-sticker vote, close the lesson by addressing the class: 'Every method you know — factorisation, completing the square, the quadratic formula — came from real people solving real problems across centuries. You are now part of that tradition. The Githurai hall is 10 metres wide and 12 metres long because someone in this room knows how to write and solve a quadratic equation.' FINAL REFLECTION: Students write independently for 3 minutes: 'In one paragraph, explain to a community elder why quadratic equations are worth learning.'	Cumulative visual model externalises the learning arc and requires synthesis rather than recall; authentic audience framing (community committee, younger students) raises the cognitive and motivational stakes; peer dot-sticker voting creates a low-stakes summative judgement that values communication as a mathematical skill; final written reflection in a community voice integrates mathematical and civic identity.	Collect the final written reflections as a lesson-exit summative artefact. Evaluate against three criteria: (1) Mathematical accuracy — does the student correctly describe what quadratic equations are and how they are solved? (2) Contextual relevance — does the student connect the mathematics to a real Kenyan scenario? (3) Justification — does the student explain why the mathematics matters, not just what it is? Use these reflections to identify students who have achieved the SLO fully, those who need targeted consolidation on specific methods, and those who show exceptional depth and may benefit from extension problems involving the discriminant or problems with no real solutions.

	member. The winning model is photographed and, if possible, displayed in the school corridor or shared with the class teacher for the end-of-term bulletin board.	Collect these as the summative written artefact of the lesson.		

D. TEACHER REFLECTION
After the lesson, consider: (1) Did all three expert groups in the Observe phase produce correct and complete solutions? If Group C (completing the square) struggled, note which step caused difficulty — moving the constant, halving the coefficient, or taking the square root — and address this in individual feedback. (2) In the Explain phase, were students able to construct the standard-form equation from a word problem independently, or did most groups need prompting to set up the quadratic before solving? If modelling was the gap, plan a short follow-up task on equation construction from word problems. (3) Did the DQB closure feel genuine to students, or did it feel procedural? If students responded with energy and ownership, the phenomenon-driven approach succeeded. If it felt routine, reflect on whether the DQB was updated consistently across all seven lessons. (4) Were the final written reflections mathematically specific or vague? Vague reflections ('quadratic equations are useful') suggest students have procedural fluency but lack conceptual articulation — a target for follow-up discussion. (5) Was 40 minutes sufficient? The Model Building phase is the most likely to overrun — if time is short, the speed gallery can be replaced with a single whole-class share from the dot-sticker-winning group. (6) Consider sending the winning model home with a note to parents explaining the Harambee hall phenomenon — this closes the community loop and reinforces the social purpose of the mathematics.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that the Githurai Harambee committee could not find the hall dimensions using arithmetic or linear equations — but when we wrote $w(w + 2) = 120$ and rearranged it to $w^2 + 2w - 120 = 0$, three different methods — factorisation, completing the square, and the quadratic formula — all gave us $w = 10$ and $w = -12$. We also observed that new community problems about fish-drying yards, tiled borders, and chapati pricing could all be modelled and solved as quadratic equations.
What did I learn?	We learned that factorisation is most efficient when the equation has integer factor pairs; completing the square always works and reveals the geometric meaning of the solution; the quadratic formula is universal and essential when the discriminant is not an obvious perfect square. We learned that both algebraic solutions must always be interpreted in context — a negative width is mathematically valid but physically meaningless. Most importantly, we learned that a quadratic equation is a precise tool for answering real planning questions that a community cannot solve by guessing.
How does this explain the phenomenon?	We explained that the three solution methods are not competing alternatives but complementary tools — each suited to different problem structures. We explained the full unit journey: quadratic expressions emerge from area and revenue contexts; expansion reveals their structure; factorisation reverses expansion to find roots; completing the square and the quadratic formula extend our reach to equations that resist factorisation. We explained to a community audience — through our models and reflections — that knowing how to write and solve a quadratic equation gives any person the power to answer real questions about land, construction, pricing, and planning in their own community.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.